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6-bar vehicle suspension linkage with drive train idler

Abstract

A two-wheel vehicle suspension linkage comprising: a suspended body **1**, a wheel carrier body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled defining a primary instantaneous velocity center at each joint between link bodies; a driven cog operatively connected to a rear wheel; a driving cog operatively connected to the suspended body **1**; an idler operatively connected by an elongated flexible member to a link body, wherein the idler is rotatable about an idler rotation axis; an elongated flexible member is disposed along a route between the driving cog and the driven cog. The idler is disposed in the route between the driving cog and the driven cog; and a damper unit is configured to resist movement between two or more of the link bodies.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
944795	12/1908	Leet et al.	N/A	N/A
5282517	12/1993	Prince	N/A	N/A
5413368	12/1994	Pong et al.	N/A	N/A
5725450	12/1997	Huskey	N/A	N/A
5901974	12/1998	Busby et al.	N/A	N/A
6203042	12/2000	Wilcox	N/A	N/A
6644677	12/2002	Rose	N/A	N/A
7059983	12/2005	Heim	N/A	N/A
7066481	12/2005	Soucek	N/A	N/A
7097190	12/2005	Matsumoto et al.	N/A	N/A
7104562	12/2005	Schmider et al.	N/A	N/A
7210695	12/2006	Griffiths	N/A	N/A
7296815	12/2006	Ellsworth et al.	N/A	N/A
7395892	12/2007	Alonzo	N/A	N/A
7556276	12/2008	Dunlap	N/A	N/A
7708297	12/2009	Thorpe	N/A	N/A
7938424	12/2010	Arraiz	N/A	N/A
7955205	12/2010	Urabe	N/A	N/A
7971892	12/2010	Sasnowski et al.	N/A	N/A
8376382	12/2012	Twers	N/A	N/A
8434776	12/2012	Wuthrich	N/A	N/A
8491429	12/2012	Cranston et al.	N/A	N/A
8991846	12/2014	Twers	N/A	N/A
9061729	12/2014	Canfield et al.	N/A	N/A
9168977	12/2014	Mcleay	N/A	N/A
9187148	12/2014	Jewell	N/A	N/A
9908583	12/2017	Matheson et al.	N/A	N/A
10093387	12/2017	Beutner	N/A	N/A
10099739	12/2017	Nishikawa	N/A	N/A
10377442	12/2018	Neilson	N/A	N/A
10464635	12/2018	Pfeiffer	N/A	N/A
10640169	12/2019	Pedretti	N/A	N/A
10703433	12/2019	Lauer	N/A	N/A
10737742	12/2019	Soncrant	N/A	N/A

10926830	12/2020	Zawistowski	N/A	N/A
11052964	12/2020	Wallace	N/A	N/A
2003/0160421	12/2002	Assier	N/A	N/A
2003/0193163	12/2002	Chamberlain et al.	N/A	N/A
2003/0193164	12/2002	Parigian	N/A	N/A
2006/0030440	12/2005	Zmurko	N/A	N/A
2006/0061059	12/2005	Lesage et al.	N/A	N/A
2006/0225942	12/2005	Weagle	N/A	N/A
2009/0261557	12/2008	Beale et al.	N/A	N/A
2009/0322055	12/2008	Arraiz	N/A	N/A
2011/0140387	12/2010	Andal et al.	N/A	N/A
2012/0142469	12/2011	Barefoot et al.	N/A	N/A
2012/0223504	12/2011	Antonot	N/A	N/A
2012/0228850	12/2011	Tseng	N/A	N/A
2014/0042726	12/2013	Canfield et al.	N/A	N/A
2014/0060950	12/2013	Beutner	N/A	N/A
2015/0035241	12/2014	McLeay	N/A	N/A
2015/0360743	12/2014	Oconnor	N/A	N/A
2016/0257371	12/2015	Droux	N/A	N/A
2016/0318582	12/2015	Johnson et al.	N/A	N/A
2016/0375956	12/2015	Talavasek et al.	N/A	N/A
2017/0101152	12/2016	Pedretti	N/A	N/A
2017/0151996	12/2016	Southall	N/A	N/A
2018/0037295	12/2017	Beale	N/A	N/A
2018/0072378	12/2017	Talavasek et al.	N/A	N/A
2018/0072379	12/2017	Talavasek et al.	N/A	N/A
2018/0072380	12/2017	Talavasek et al.	N/A	N/A
2018/0072739	12/2017	Goldman et al.	N/A	N/A
2018/0140387	12/2017	Richard	N/A	N/A
2018/0148123	12/2017	Neilson	N/A	B62K 25/286
2018/0265165	12/2017	Zawistowski	N/A	N/A
2018/0304958	12/2017	Madsen	N/A	N/A
2019/0039682	12/2018	Zawistowski	N/A	N/A
2019/0300097	12/2018	Chamberlain	N/A	N/A
2020/0031428	12/2019	Lund et al.	N/A	N/A
2020/0079470	12/2019	Saiki	N/A	N/A
2020/0247500	12/2019	Zawistowski	N/A	N/A
2021/0046996	12/2020	Beale	N/A	N/A
2022/0153381	12/2021	Zawistowski et al.	N/A	N/A
2022/0306240	12/2021	Talavasek et al.	N/A	N/A
2023/0070334	12/2022	Zawistowski	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2016102059	12/2016	AU	N/A
19726067	12/2005	DE	N/A
102004059187	12/2005	DE	N/A
102019002456	12/2018	DE	N/A
1026073	12/1999	EP	N/A
1238900	12/2001	EP	N/A

2706002	12/2013	EP	N/A
1799534	12/2013	EP	N/A
2812234	12/2013	EP	N/A
3235711	12/2016	EP	N/A
2520321	12/1982	FR	N/A
2766452	12/1998	FR	N/A
2807730	12/2001	FR	N/A
1087559	12/1966	GB	N/A
2086319	12/1981	GB	N/A
2338216	12/1998	GB	N/A
2522461	12/2014	GB	N/A
2590808	12/2021	GB	N/A
2605244	12/2021	GB	N/A
2027223	12/2021	NL	N/A
03064243	12/2002	WO	N/A
2010143795	12/2009	WO	N/A
2015004490	12/2014	WO	N/A
2018170505	12/2017	WO	N/A
2019010394	12/2018	WO	N/A
2020107055	12/2019	WO	N/A
2021133996	12/2020	WO	N/A
2021174088	12/2020	WO	N/A
2021257865	12/2020	WO	N/A
2023039543	12/2022	WO	N/A

OTHER PUBLICATIONS

U.S. Patent and Trademark Office, Final Office Action mailed on Jul. 22, 2021, for U.S. Appl. No. 15/925,165, 15 pages. cited by applicant

European Patent Office, EP Extended Search Report, mailed Sep. 21, 2022, for European Application No. 20747607.8, 2 pages. cited by applicant

Netherlands Intellectual Property Office, Search Report and Opinion, issued Nov. 15, 2021, for Dutch Application No. 2027668, 24 pages. cited by applicant

Smurthwaite, James, Pinkbike, "Spotted: A New Commencal Supreme Breaks Cover at the Portugal Cup," published Mar. 7, 2022, 4 pages. cited by applicant

Roberts, Dan, Pinkbike, "What's Going On With Commencal's Prototype DH Race Bike?", published Jun. 9, 2021, 9 pages. cited by applicant

European Patent Office, "Communication Pursuant to Article 94(3) EPC," mailed Jul. 8, 2022, for European Application No. 18768549.0, 6 pages. cited by applicant

European Patent Office, "Extended European Search Report," mailed Oct. 4, 2022, for European Application No. 20747607.8, 6 pages. cited by applicant

German Patent Office, "Examination Notice and Search Report," mailed Apr. 13, 2022, for German Application No. 10 2020 134 843.6, 7 pages. cited by applicant

U.K. Intellectual Property Office, "Search Report," mailed Jul. 7, 2022, for U.K. Application No. GB2208682.1, 3 pages. cited by applicant

U.S. Patent and Trademark Office, "Non-Final Office Action," mailed Sep. 16, 2022, for U.S. Appl. No. 16/705,049, 10 pages. cited by applicant

Brown, "Preview: Nicolai Bikes Available in the U.S.: The belt-drive, big-hit bikes will be distributed by Nicolai USA," Bike Mag, Jun. 3, 2014, 9 pages. cited by applicant

Overholt, "Soc 14: Effigear Calls in the Cavalerie for new Gear Box Bikes in the US," Bike Rumor, Apr. 23, 2014, 16 pages. cited by applicant

U.S. Patent and Trademark Office, “International Search Report,” mailed Jan. 20, 2023, for PCT Application No. PCT/US2022/076232, 2 pages. cited by applicant

U.S. Patent and Trademark Office, “Written Opinion,” mailed Jan. 20, 2023, for PCT Application No. PCT/US2022/076232, 11 pages. cited by applicant

“Combined Search and Examination Report under Sections 17 and 18(3)”, mailed by U.K. Intellectual Property Office on Aug. 25, 2021, for U.K. Application No. GB2102854.3, 8 pages. cited by applicant

“Extended European Search Report for EP 18768549.0”, mailed Feb. 8, 2021. cited by applicant

“International Search Report & Written Opinion”, mailed on Jul. 16, 2021, for PCT Application No. PCT/US2021/020034, 16 pages. cited by applicant

“International Search Report & Written Opinion mailed Apr. 13, 2021, in PCT Application No. PCT/US2020/066980, 16 pages”. cited by applicant

“Netherlands Patent Office, Written Opinion and Search Report mailed Sep. 24, 2021”, in Dutch and English, for Netherlands Application No. 2027223, 17 pages. cited by applicant

“U.K. Intellectual Property Office, “Combined Search and Examination Report under Sections 17 and 18(3),””, mailed Apr. 21, 2021, for U.K. Application No. GB2020235.4, 6 pages. cited by applicant

GB IPO, “Combined Search and Exam Report”, App. No. 2020235.4, Apr. 21, 2021, 6 pages. cited by applicant

MTBR: Mountain Bike Review Forum, , “Jayem Discussion Starter #1—Knolly Suing Intense for Building Bikes with Seat-tubes in Front of the BB”, <https://www.mtbr.com/threads/knolly-suing-intense-for-building-bikes-with-seat-tubes-in-front-of-the-bb.1173867/>, Jan. 2021, 16 pages. cited by applicant

Worsey, “Forbidden Druid Review—Are high pivots just for downhill?”, Enduro, [online], Mar. 4, 2019 [retrieved on Jun. 16, 2021], From Internet: <url: <https://enduro-mtb.com/en/fobidden-druid-review/>>; 15 pages. cited by applicant

Netherlands Patent Office, “Written Opinion and Search Report,” mailed Feb. 22, 2023, for Dutch Application No. 2029897, Dutch with partial English translation, 12 pages. cited by applicant

Aston, P., “Robot Bike Co R160 Custom—Review,” Outside Magazine, Oct. 17, 2016, 46 pages. cited by applicant

Nicolai Maschinenbau, “History of Nicolai,” Gesellschaft für Zweirad und Maschinenbau mbh, at least as early as 1995, 3 pages. cited by applicant

Nicolai, “Nicolai Trombone Frame,” at least as early as 1995, 6 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority pursuant to 35 U.S.C. § 119 (e) of U.S. provisional patent application No. 62/983,322, filed 28 Feb. 2020 and entitled “6-Bar Vehicle Suspension Linkage with Drive Train Idler,” which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) The technology described herein relates to vehicle suspension systems, specifically, to linkages and drive trains within a vehicle suspension system.

BACKGROUND

(2) Vehicle suspension terminology depends upon the reference frame considered. Consider a static vehicle that has two wheels, each of which are supported by the ground. Such vehicles can include a suspended body and a non-suspended body operatively coupled to each wheel. In a two-wheel vehicle, such as a bicycle, electric bicycle or pedelec or motorcycle, etc. there is typically one rear wheel known as the driven wheel, which includes a driven cog. The driven cog is often part of a group of cogs called a cassette wherein different cog sizes may be shifted into to alter the gear ratio. There is also one front wheel. The driving cog is typically coupled to the suspended body. A derailleur mechanism may be used to shift the driving chain or belt into various driven cogs of the cassette. The driving cog, which is connected to the driven cog via the driving chain/belt, is rotated by a crank under human power, or by a motor, or by combined motor and human power. The reaction of the driven wheel and the ground causes the vehicle to accelerate forward, or in the general direction from the rear wheel to the front wheel. Rearward is then defined as the general direction from the front wheel to the rear wheel.

(3) A linkage operatively couples the suspended body and the driven wheel. A linkage may be composed of multiple bodies (often referred to as links or members) that are typically coupled to each other in a manner that allows the bodies the suspended body and the non-suspended body to move relative to one another, such as by flexing, camming, rotating, and/or translating relative to one another. The linkage constrains the movement of the suspended body and the non-suspended body relative to one another, during which movement of the driven wheel and brake on the non-suspended body may move relative to the suspended body. A combination of damper(s) and/or spring(s) is/are typically arranged to react to relative motion between the suspended body and the driven wheel. The linkage may be highly responsible for the vehicle's dynamic response to acceleration and deceleration as well as the mechanical advantage over the shock/damper. While the shock/damper may resist movement of the suspension linkage, the path of the driven wheel and/or driven wheel axis between extended and compressed positions of the suspension linkage is not affected by the presence or absence of the shock/damper. The shock/damper does not affect the degrees of freedom of the suspension linkage. The shock/damper does not affect the relative motion of the links making up the suspension linkage is not affected.

(4) The information included in this Background section of the specification, including any references cited herein and any description or discussion thereof, is included for technical reference purposes and is not to be regarded subject matter by which the scope of the invention as defined in the claims is to be bound.

SUMMARY

(5) The technology disclosed herein relates to vehicle suspension linkages. In one embodiment, a two-wheel vehicle suspension linkage is provided. The suspension includes a suspended body **1**, a link body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled with one another and a idler cog **56** with rotation axis **75** that is pivotally connected to suspended body **1**. Link body **2** is the wheel carrier and the brake carrier body in this embodiment. The link body **3** includes jointed connections with the suspended body **1** defining PIVC[1][3], the link body **4** defining PIVC[3][4], and the link body **6** defining PIVC[3][6]. Idler cog rotation axis **75** is coincident to PIVC[1][3]. The link body **4** includes an additional jointed connection with the link body **2** defining PIVC[2][4]. The link body **5** includes additional jointed connections with suspended body **1** defining PIVC[1][5], link body **2** defining PIVC[2][5], and the link body **6** defining PIVC[5][6]. The suspension may be coupled to a damper unit configured to resist movement between two or more of the suspended body **1**, link body **2**, link body **3**, link body **4**, link body **5**, or link body **6**. The damper unit may include an extension body or bodies to increase its effective length.

(6) In yet another embodiment, a two-wheel vehicle suspension linkage is provided. The suspension includes a suspended body **1**, a swingarm body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled with one another and a idler cog **56** with rotation

axis 75 that is pivotally connected to suspended body 1. In various embodiments, the link body 2 is the wheel carrier and the brake carrier body. The link body 3 includes jointed connections with the suspended body 1 defining PIVC[1][3], the link body 4 defining PIVC[3][4], and the link body 6 defining PIVC[3][6]. Idler cog rotation axis 75 is not coincident to PIVC[1][3]. The link body 4 includes an additional jointed connection with the link body 2 defining PIVC[2][4]. The link body 5 includes additional jointed connections with suspended body 1 defining PIVC[1][5], link body 2 defining PIVC[2][5], and the link body 6 defining PIVC[5][6]. The suspension may be coupled to a damper unit configured to resist movement between two or more of the suspended body 1, link body 2, link body 3, link body 4, link body 5, or link body 6. The damper unit may include an extension body or bodies to increase its effective length.

(7) In one aspect, a two-wheel vehicle suspension linkage includes: a suspended body 1, a wheel carrier body 2, a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled defining a primary instantaneous velocity center (PIVC) at each joint between link bodies; a wheel operatively connected to the rear wheel carrier body 2; a driven cog operatively connected to the rear wheel; a driving cog operatively connected to the suspended body 1; an idler operatively connected by an elongated flexible member to the suspended body 1, the wheel carrier body 2, the link body 3, the link body 4, the link body 5 or the link body 6, where the idler is rotatable about an idler rotation axis; an elongated flexible member disposed along a route between the driving cog and the driven cog, where the idler is disposed in the route between the driving cog and the driven cog; and a damper unit configured to resist movement between two or more of the suspended body 1, wheel carrier body 2, link body 3, link body 4, link body 5, or link body 6.

(8) The two-wheel vehicle suspension linkage may also include where the idler is pivotally connected to the suspended body 1.

(9) The two-wheel vehicle suspension linkage of any may also include, where the idler is rotatably connected to suspended body 1 concentrically with a linkage pivot with the wheel carrier body 2, the link body 3, the link body 4, the link body 5 or the link body 6.

(10) The two-wheel vehicle suspension linkage may also include where the idler rotation axis is coincident to a PIVC defined by a combination of the suspended body 1, the wheel carrier body 2, the link body 3, the link body 4, the link body 5, or the link body 6.

(11) The two-wheel vehicle suspension linkage may also include where the link body 3 has jointed connections with the suspended body 1 defining a PIVC[1][3], the link body 4 defining a PIVC[3][4], and the link body 6 defining a PIVC[3][6]. The two-wheel vehicle suspension linkage may also include the link body 4 with a jointed connection with wheel carrier body 2 defining a PIVC[2][4]. The two-wheel vehicle suspension linkage may also include the link body 5 with jointed connections with the suspended body 1 defining a PIVC[1][5], the wheel carrier body 2 defining a PIVC[2][5], and the link body 6 defining a PIVC[5][6], where the idler rotation axis is not coincident to the PIVC[1][3].

(12) The two-wheel vehicle suspension linkage may also include where the idler is an idler cog.

(13) The two-wheel vehicle suspension linkage may also include where at least one PIVC migration path reverses as the suspension linkage moves from an at least partially extended state to an at least partially compressed state.

(14) The two-wheel vehicle suspension linkage may also include where the suspended body 1 has a motive power source compartment, and an energy storage compartment selectively coverable by a panel.

(15) The two-wheel vehicle suspension linkage may also include where at least one PIVC migration path reverses as the suspension linkage moves from an at least partially extended state to an at least partially compressed state.

(16) The two-wheel vehicle suspension linkage may also include where the driving cog axis is rotatable about a driving cog axis and the idler rotation axis is disposed above and forward of the driving cog axis.

(17) The two-wheel vehicle suspension linkage may also include where a contact angle formed between a forward vertical tangent of the driving cog and line that is tangent to both the driving cog and the idler is greater than zero degrees.

(18) The two-wheel vehicle suspension linkage may also include an instantaneous force center (IFC) located at an intersection of a force vector of the flexible member and a driving force vector, where, as the suspension linkage moves between an extended and compressed state, the IFC moves from an extended position rearward to an intermediate position and then forward to compressed position.

(19) The two-wheel vehicle suspension linkage may also include where an anti-squat percentage of the suspension linkage increases as a vertical travel of the wheel increases to a first position, where, as the vertical wheel travel increases further to a second position, the anti-squat percentage falls.

(20) The two-wheel vehicle suspension linkage may also include where a rate of change of a power transmitting length of the flexible member (dPTL) increases as a vertical travel of the wheel increases to a first position, where, as the vertical wheel travel increases further to a second position, the dPTL falls.

(21) The two-wheel vehicle suspension linkage may also include where the idler is mounted to the suspended body **1**. Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(22) In one aspect, a two-wheel vehicle suspension linkage includes a suspended body **1**, and a plurality of link bodies including a wheel carrier body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled defining a primary instantaneous velocity center (PIVC) at each joint between the plurality of link bodies. The two-wheel vehicle suspension linkage also includes a driving cog having a driving cog diameter and operatively connected to the suspended body **1**, where the driving cog is rotatable in a driving cog center plane about a driving cog axis normal to the driving cog center plane, and a driving cog region is defined between a front driving cog plane tangent to a front of the driving cog diameter and parallel to the driving cog axis, and a rear driving cog plane tangent to a rear of the driving cog diameter and parallel to the driving cog axis, where at least a portion of one or more of the plurality of link bodies is located inside the driving cog region and extends transversely outward past the driving cog center plane.

(23) In one aspect, a two-wheel vehicle suspension linkage includes a suspended body **1**, and a plurality of link bodies including a wheel carrier body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled defining a primary instantaneous velocity center (PIVC) at each joint between the plurality of link bodies. The two-wheel vehicle suspension linkage also includes a driving cog operatively connected to the suspended body **1** defining a driving cog axis of rotation and a driving cog radius; where the link body **3** includes jointed connections with the suspended body **1** defining a PIVC[1][3], the link body **4** defining a PIVC[3][4], and the link body **6** defining a PIVC[3][6]. The two-wheel vehicle suspension linkage also includes the link body **4** has a jointed connection with wheel carrier body **2** defining a PIVC[2][4]. The two-wheel vehicle suspension linkage also includes the link body **5** has jointed connections with the suspended body **1** defining a PIVC[1][5], the wheel carrier body **2** defining PIVC[2][5], and the link body **6** defining an PIVC[5][6], where the driving cog radius is about 40%-65% of a distance from the PIVC [1][3] to the driving cog axis of rotation.

(24) In one aspect, a two-wheel vehicle suspension linkage includes a suspended body **1**, and a plurality of link bodies including a wheel carrier body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled defining a primary instantaneous velocity center (PIVC) at each joint between the plurality of link bodies. The two-wheel vehicle suspension linkage also includes a driving cog operatively connected to suspended body **1** defining a driving cog radius; where the link body **3** includes jointed connections with the suspended body **1** defining a PIVC[1][3], the link body **4** defining a PIVC[3][4], and the link body **6** defining a PIVC[3][6]. The two-wheel vehicle suspension linkage also includes the link body **4** has a jointed connection with wheel

carrier body 2 defining a PIVC[2][4]. The two-wheel vehicle suspension linkage also includes the link body 5 has jointed connections with the suspended body 1 defining a PIVC[1][5], the wheel carrier body 2 defining a PIVC[2][5], and the link body 6 defining a PIVC[5][6], where the driving cog radius is about 40%-70% of a distance from the PIVC[3][6] to the PIVC[5][6].

(25) In one aspect, a two-wheel vehicle suspension linkage includes a suspended body 1, a wheel carrier body 2, a wheel operatively connected to the rear wheel carrier body 2, a driven cog operatively connected to the rear wheel, a driving cog operatively connected to the suspended body 1, an idler operatively connected by an elongated flexible member to the suspended body 1, where the idler is rotatable about an idler rotation axis, and an elongated flexible member disposed along a route between the driving cog and the driven cog, where the idler is disposed in the route between the driving cog and the driven cog, where an instantaneous force center (IFC) is located at an intersection of a force vector of the flexible member and a driving force vector, where, as the suspension linkage moves between an extended and compressed state, the IFC moves from an extended position rearward to an intermediate position and then forward to compressed position.

(26) The two-wheel vehicle suspension linkage may also include a damper unit configured to resist movement between two or more of the suspended body 1, wheel carrier body 2, link body 3, link body 4, link body 5, or link body 6 Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(27) In one aspect, a two-wheel vehicle suspension linkage includes a suspended body 1, a wheel carrier body 2, a wheel operatively connected to the rear wheel carrier body 2, a driven cog operatively connected to the rear wheel, a driving cog operatively connected to the suspended body 1, an idler operatively connected by an elongated flexible member to the suspended body 1, where the idler is rotatable about an idler rotation axis, and an elongated flexible member disposed along a route between the driving cog and the driven cog, where the idler is disposed in the route between the driving cog and the driven cog, where a rate of change of a power transmitting length of the flexible member (dPTL) increases as a vertical travel of the wheel increases to a first position, where, as the vertical wheel travel increases further to a second position, the dPTL falls.

(28) The two-wheel vehicle suspension linkage may also include a damper unit configured to resist movement between two or more of the suspended body 1, wheel carrier body 2, link body 3, link body 4, link body 5, or link body 6 Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(29) The two-wheel vehicle suspension linkage may also include where the link body 3 has jointed connections with the suspended body 1 defining a PIVC[1][3], the link body 4 defining a PIVC[3][4], and the link body 6 defining a PIVC[3][6]. The two-wheel may also include the link body 4 has a jointed connection with the wheel carrier body 2 defining a PIVC[2][4]. The two-wheel may also include the link body 5 has jointed connections with the suspended body 1 defining a PIVC[1][5], the wheel carrier body 2 defining a PIVC[2][5], and the link body 6 defining a PIVC[5][6]; where the idler rotation axis is coincident to the PIVC[1][3].

(30) The two-wheel vehicle suspension linkage may also include where the idler is mounted to an axle, where the axle is pivotally connecting a wheel carrier body 2, linkage body 3, linkage body 4, linkage body 5, or linkage body 6 to suspended body-1.

(31) The two-wheel vehicle suspension linkage may also include where the link body 6 defines a longitudinal axis, and the longitudinal axis moves from a position behind the PIVC[1][5] when the suspension linkage is in an at least partially extended state to a position in front of the PIVC[1][5] when the suspension linkage is in an at least partially compressed state.

(32) The two-wheel vehicle suspension linkage may also include a motive power source received in the motive power source compartment, an energy storage module received in the energy storage compartment and contained by the panel, an electrical conduit in electrical communication with the motive power source and the energy storage module and operative to supply electrical power from the energy storage module to the motive power source.

- (33) The two-wheel vehicle suspension linkage may also include where the idler rotation axis is coincident with a PIVC.
- (34) The two-wheel vehicle suspension linkage may also include where the idler rotation axis is not coincident with a PIVC.
- (35) The two-wheel vehicle suspension linkage may also include where the compressed position is forward of at least one of the extended position or the intermediate position.
- (36) The two-wheel vehicle suspension linkage may also include where the dPTL is maximized near a sag point of the rear suspension.
- (37) The two-wheel vehicle suspension linkage may also include where the dPTL is maximized at a first value of the vertical travel of the wheel, and where the link body 3 has an inflection point at a second value of the vertical travel of the wheel greater than the first value.
- (38) The two-wheel vehicle suspension linkage may also include a damper unit configured to resist movement between two or more of the suspended body 1, the wheel carrier body 2, the link body 3, the link body 4, the link body 5, or the link body 6.
- (39) The two-wheel vehicle suspension linkage may also include a damper unit configured to resist movement between two or more of the suspended body 1, the wheel carrier body 2, the link body 3, the link body 4, the link body 5, or the link body 6.
- (40) The two-wheel vehicle suspension linkage may also include a damper unit configured to resist movement between two or more of the suspended body 1, the wheel carrier body 2, the link body 3, the link body 4, the link body 5, or the link body 6.
- (41) The two-wheel vehicle suspension linkage may also include a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled with one another and the suspended body 1 and wheel carrier body 2 and defining a primary instantaneous velocity center (PIVC) at each joint between link bodies.
- (42) The two-wheel vehicle suspension linkage may also include a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled with one another and the suspended body 1 and wheel carrier body 2 and defining a primary instantaneous velocity center (PIVC) at each joint between link bodies. Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.
- (43) The two-wheel vehicle suspension linkage may also include where the PIVC [1][3] 7 is spaced apart from the idler rotation axis by a distance of less than or equal to 50-mm.
- (44) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. A more extensive presentation of features, details, utilities, and advantages of the present invention as defined in the claims is provided in the following written description of various embodiments and implementations and illustrated in the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A shows a 6-bar suspension linkage system with a typical configuration in the extended state.
- (2) FIG. 1B is a section of the embodiment of FIG. 1A with cross section C-C.
- (3) FIG. 1C shows cross-section C-C of the embodiment of FIG. 1B.
- (4) FIG. 2A shows an embodiment of a 6-bar suspension linkage system in the extended state.
- (5) FIG. 2B shows the embodiment of FIG. 2A of a bicycle with the disclosed 6-bar suspension linkage system in the extended state.

- (6) FIG. 2B-1 shows a partial enlarged view of the embodiment of FIG. 2B.
- (7) FIG. 2C shows the embodiment of FIG. 2A of a 6-bar suspension linkage system in the compressed state.
- (8) FIG. 2D shows the embodiment of FIG. 2A of a bicycle with the disclosed 6-bar suspension linkage system in the compressed state.
- (9) FIG. 2D-1 shows a partial enlarged view of the embodiment of FIG. 2B.
- (10) FIG. 2E shows the embodiment of FIG. 2A of a 6-bar suspension linkage system in the extended state with cross-section A.
- (11) FIG. 2F shows cross-section A-A of the embodiment of FIG. 2A.
- (12) FIG. 2G is a section of the embodiment of FIG. 2B with cross section D-D.
- (13) FIG. 2H shows cross-section D-D of the embodiment of FIG. 2G.
- (14) FIG. 3A shows another embodiment of a 6-bar suspension linkage system in the extended state.
- (15) FIG. 3B shows the embodiment of FIG. 3B of a bicycle with the disclosed 6-bar suspension linkage system in the extended state.
- (16) FIG. 3C shows the embodiment of FIG. 3B of a 6-bar suspension linkage system in the compressed state.
- (17) FIG. 3D shows the embodiment of FIG. 3B of a bicycle with the disclosed 6-bar suspension linkage system in the compressed state.
- (18) FIG. 3E shows the embodiment of FIG. 3B of a 6-bar suspension linkage system in the extended state with cross-section B.
- (19) FIG. 3F shows cross-section B-B of the embodiment of FIG. 3B.
- (20) FIG. 4 is a schematic illustrating an anti-squat and anti-rise analysis of a 6-bar suspension with an idler.
- (21) FIG. 5 is a graph depicting expected/exemplary anti-squat curves of a 6-bar suspension linkage system with a typical configuration and an expected/exemplary anti-squat curve of the embodiments illustrated in FIGS. 2A and 3A.
- (22) FIG. 6 is a graph of an expected/exemplary anti-rise curve of the embodiments illustrated in FIGS. 2A and 3A.
- (23) FIG. 7 is a graph of an expected/exemplary leverage rate curve of the embodiments illustrated in FIGS. 2A and 3A.
- (24) FIG. 8A is an exploded isometric view of a 6-bar suspension linkage for an electric bike.
- (25) FIG. 8B is a partial exploded elevation view of the suspension linkage of FIG. 8A.
- (26) FIG. 8C is a partial right elevation view of the suspension linkage of FIG. 8A.
- (27) FIG. 8D is a partial right elevation view of the suspended body of the suspension linkage of FIG. 8A.
- (28) FIG. 8E is a partial left elevation view of the suspension linkage of FIG. 8A.
- (29) FIG. 8F is a partial right elevation view of the suspension linkage of FIG. 8A in an extended state.
- (30) FIG. 8G is a partial right elevation view of the suspension linkage of FIG. 8A at an intermediate state between an extended and compressed state.
- (31) FIG. 8H is a partial right elevation view of the suspension linkage of FIG. 8A in a compressed state.
- (32) FIG. 8I is a partial right elevation view of the suspension linkage of FIG. 8A showing a contact angle of the belt/chain to the driving cog.
- (33) FIG. 9A is an exploded isometric view of a 6-bar suspension linkage for an electric bike.
- (34) FIG. 9B is a partial exploded elevation view of the suspension linkage of FIG. 9A.
- (35) FIG. 9C is a partial right elevation view of the suspension linkage of FIG. 9A.
- (36) FIG. 9D is a partial left elevation view of the suspension linkage of FIG. 9A.
- (37) FIG. 9E is a partial right elevation view of the suspension linkage of FIG. 9A in an extended

state.

(38) FIG. 9F is a partial right elevation view of the suspension linkage of FIG. 9A at an intermediate state between an extended and compressed state.

(39) FIG. 9G is a partial right elevation view of the suspension linkage of FIG. 9A in a compressed state.

(40) FIG. 10 is a partial isometric view of an embodiment of a suspension linkage.

(41) FIG. 11 is a partial exploded isometric view of the suspension linkage of FIG. 10.

(42) FIG. 12 is a partial elevation view of the suspension linkage of FIG. 10.

(43) FIG. 13 is a partial section view of the suspension linkage of FIG. 10 taken along section line E-E of FIG. 12.

(44) FIG. 14 shows a selected portion of the 15 IVC migration paths of the embodiments shown in FIGS. 1A, 2A, 3A, 8A, and 9A.

(45) FIG. 15 shows an exploded isometric view of a portion of a suspension linkage for an electric bike.

(46) FIG. 16 shows a partial elevation view of a portion of the suspension linkage of the FIG. 15.

(47) FIG. 17A shows an example of a key suitable for use with the suspension linkages disclosed herein.

(48) FIG. 17B shows a side elevation view of the key of FIG. 17A.

(49) FIG. 17C shows a section view of the key of FIG. 17A taken along section line 17C-17C of FIG. 17B.

(50) FIG. 18 shows a partial section view of the portion of the suspension linkage of FIG. 16 taken along section line 18-18 of FIG. 16.

(51) FIG. 19 is a graph depicting expected/exemplary anti-squat curves of a 6-bar suspension linkage system of the present disclosure with relatively smaller and larger driven mechanisms (e.g., differently sized cogs of a rear cassette).

(52) FIG. 20 is a graph depicting an expected/exemplary migration path of an instantaneous force center as a suspension linkage moves between extended and compressed states.

(53) FIG. 21A shows an example of a suspension linkage of an electric bike in an at least partially extended position.

(54) FIG. 21B shows the suspension linkage of FIG. 22A in an at least partially contracted position.

(55) FIG. 22 is a graph depicting an expected/exemplary rate of change of the power transmitting length of the suspension linkages of FIGS. 1A, 2A, 3A, 8A, 9A, 14, and/or 18A as it relates to vertical wheel travel.

DETAILED DESCRIPTION

(56) Disclosed herein is a system or linkage that operatively couples a suspended body to a driven wheel. The suspended body and non-suspended body may move relative to one another, as defined by the system or linkage that operably couples them. In some embodiments, the driven wheel is associated with the non-suspended body (e.g., a rear triangle, chain stay and/or seat stay). In some embodiments, the driven wheel may be associated with the suspended body (e.g., a front wheel drive vehicle). In accordance with the various embodiments provided herein, the suspension system linkage improves suspension performance based on the interrelationships of its linkage bodies and the related instantaneous velocity centers (IVCs). In one example, the linkage has 15 IVCs. In particular, the linkage may be a 6-bar linkage. The 6-bar linkage may define a suspended body portion and a non-suspended body portion. Moreover, an additional idler may be operatively coupled to the suspended body or the non-suspended body. In this case, a driving chain or belt is routed in a loop around the driven mechanism (e.g. driven cog), the idler and the driving mechanism (e.g. driving cog). Either or both of the driven mechanism or the driving mechanism may be a respective single member (e.g., a single cog, pulley, or ring), or the driving and driven mechanisms may include a cassette with a plurality of members (e.g., cogs, pulleys, or rings). The idler may be located between the driving mechanism and the driven mechanism so that the driving

mechanism transmits the force to the driven mechanism via the idler. In various embodiments, the idler is specifically located with respect to physical IVCs, or PIVCs that are unique to a 6-bar linkage.

(57) Various theories, methods, algorithms or analysis systems are provided herein. These systems are provided for better understanding of the structures and configurations described. Unless specifically claimed, the systems are not limiting regardless of current accuracy or subsequent clarifications or understandings of the structures and configurations that may be determined by persons of ordinary skill in the art.

(58) Accordingly, the positions of IVCs may change depending on the configuration of the system. A particular IVC may be in a different position depending on whether the system is in a compressed or extended state defining an IVC migration path. Thus, the interrelationships between IVCs can be analyzed, allowing for a mathematical analysis of the movement of the linkage subsystem. Additionally, methods to calculate anti-squat, anti-rise, and leverage rate is detailed. While support is provided herein, Pat. Pub. No. US2018-0265165A1 provides additional detailed support of these methods, algorithms and analysis systems and the 6-bar linkage suspension and is hereby incorporated by reference in its entirety.

(59) It is understood that throughout this disclosure the relationship of various linkages are described with respect to characteristics and structures of those linkages. One analysis system useful for assessing these relationships is the Mobility Analysis of Mechanisms. The Mobility Analysis of Mechanisms (Kutzbach (or Grübler) mobility criterion) may be used to describe the mobility, or output degree of freedom of a linkage. This system may be used to describe the mobility m of a planar linkage composed of n links that are coupled with p flexible joints/pivots. In the various embodiments, discussed herein, the links can be connected via these flexible joints/pivots allowing some degree of freedom between one another. Additionally, the relationship of the links via the joints define various characteristics such as instantaneous velocity centers (IVCs). In various examples as applied to the various embodiments discussed herein, the flexible joints/pivots can include revolute, slider, cam joints, or any other suitable flexible joints or pivots that allow one degree of freedom movement between the two links they connect. Notably, flexible joints may include intermediary devices connecting the linkages. Depending on the types of joints, quality of joints, or the tolerances in the joints, characteristics (e.g. the IVCs or other characteristics discussed herein) may have small variances between joints due to real world engineering constraints and calculations. Terminology such as generally, substantially, or other similar terms may be used to account for the expected, calculated, or otherwise real-world accuracy of the characteristics discussed herein while allowing for real world variance in the characteristics. Note that if bodies are coupled as one and not considered rigid, a specific joint (e.g. a revolute joint) may be assumed theoretically for analysis near the point of flexure in the flexible joint. Also, note that although the linkage is considered planar kinematically, the assembly of the mechanism may be 3-dimensional.

(60) The following equation is used for analysis of the various systems herein:

$\text{mobility} = m = 3(n - 1 - p) + p$ n =number of bodies (or links or members) p =number of joints Σf =sum of the kinetic variables in the mechanism

(61) As an example, this equation may be applied to a 4-bar linkage. The following solves the equation for a 4-bar linkage:

$$p = n = 4$$

$$m = 3(n - 1 - p) + p$$

$$m = 3(4 - 1 - 4) + 4$$

$$m = 3(-1) + 4$$

$$m = -3 + 4$$

$$m = 1$$

(62) As another example, this equation may be applied to a 6-bar linkage. The following solves the

equation for a 6-bar linkage:

$$n=6$$

$$p=7$$

$$m=3(n-1-p)+p$$

$$m=3(6-1-7)+7$$

$$m=3(-2)+7$$

$$m=-6+7$$

$$m=1$$

(63) In both noted 4-bar and 6-bar linkages, $m=1$, or there is one degree of freedom of motion. Therefore, the path of the axis of the driven wheel, known as the driven wheel axis path (DWAP) may be constrained to planar motion along a defined path or curve relative to the suspended body. This path or curve includes one end-point defined as the extended state, and another end-point as the compressed state. Any point on this curve or path between the extended and compressed points is known as an intermediate state. An intermediate state on an IVC migration curve or path correlates to an intermediate state of the linkage positions.

(64) Additionally, methods of analyzing vehicle suspension linkages design for its dynamic response is also disclosed. In one example, this method of analysis includes a collection of the system instantaneous velocity centers (IVCs), which can be determined graphically. An IVC is a point common to two linkage bodies where there is zero relative velocity. These IVCs change location instantaneously as the suspension is cycled from its extended to compressed state. The path of each IVC migration may then be plotted graphically as a path, curve, or spline from the extended to the compressed state. These IVC curves depend upon the reference frame considered. In various embodiments, the suspended body is considered fixed as the driven wheel moves from the extended to the compressed state. Total suspension travel (VWT [T]) is then defined as the perpendicular distance relative to the ground line at the extended state as measured between the extended suspension state point and the compressed suspension state point on the driven wheel axis path.

(65) It is possible for an IVC, known as a stationary IVC (SIVC), to have little to no migration from the extended to the compressed state. One example would be an IVC where a link body is operatively connected to the suspended body. This is a result of the front-triangle remaining fixed in the reference frame chosen for suspension analysis.

(66) For reference herein, specific instantaneous velocity centers of a linkage are denoted as IVC[Body-A][Body-B]. Body-A and Body-B being the relevant bodies in the relationship. For example, IVC[1][2] is the instantaneous velocity center relative to a body 1 and a body 2. Additionally, IVC[1][2] is equivalent to IVC[2][1].

(67) The structure surrounding the suspension system may include several bodies. In various examples, the structure may include a suspended body. In various embodiments, the suspended body can be suitable to be supported by suspension and support a user over the suspension. In various examples, the structure may include a “wheel carrier” body, which is operatively coupled to the driven wheel, a “brake carrier” body, which is operatively coupled to the driven wheel brake, or a “dynamic body” (DB), which is any combination of a wheel carrier and a brake carrier body (e.g., DB=wheel carrier body, or DB=brake carrier body, or DB=wheel and brake carrier body).

(68) Specific IVC migrations called dynamic IVCs (DIVCs) may be utilized to determine the vehicle's dynamic response. The DIVCs depend upon the specific linkage layout but also depend upon suspended body-1 since this is the body in which a passenger or rider will be included. Suspended body-1 may be referenced herein as the front triangle of a bicycle.

(69) As used herein DIVC[AD] can define both the acceleration and deceleration response of the vehicle; DIVC[A] can define the acceleration response of the vehicle; and DIVC[D] can define the deceleration response of the vehicle. As used herein DIVC[C] is defined as a DIVC that includes an acceleration component. As such, in certain embodiments, DIVC[C] can be equal to DIVC[A] or

DIVC[AD]. As used herein DIVC[E] is defined as a DIVC that includes a deceleration component. As such, in certain embodiments, DIVC[E] can be equal to DIVC[D] or DIVC[AD].

(70) As used herein DIVC is a general term that may generically cover a DIVC[AD] or a DIVC[A] or a DIVC[D] or a DIVC[C] or a DIVC[E] or any combination thereof. As used herein DIVC[L] is the length of the DIVC migration path, spline or curve.

(71) In accordance with various embodiments, the body-X can be both a wheel carrier and a brake carrier body. In such an embodiment, there is a single DIVC[AD] migration, DIVC[AD][1][X].

(72) In accordance with various embodiments, wheel carrier body-Y can be separate from the brake carrier body-Z. In such an embodiment, there are two DIVCs, DIVC[A][1][Y] and DIVC[D][1][Z].

(73) Finally, in accordance with various embodiments, the wheel carrier body-Y is pivotally concentric to a brake carrier body-Z. In this case, again there are two DIVCs, DIVC[A][1][Y] and DIVC[D][1][Z].

(74) Each of these various embodiments can be variously applied to the embodiments and examples of the various systems discussed in more detail below.

(75) For purposes of understanding, but not to be limiting, it can be noted that the point in which the force of gravity acts on the sum of the suspended vehicle mass (also known as the sprung mass) and any additional mass such as a passenger or cargo that is supported by the suspension is known as the center of gravity (COG). In the static case, with both wheels on the ground, the force due to the suspended mass through the COG is supported by the vehicle's two wheels. Depending on the COG location and the wheelbase of the vehicle, the distribution of force between the two wheels may vary. When the vehicle accelerates, load transfer occurs and the force distribution between the two wheels changes. The rear wheel load is increased while the front wheel load is decreased. Thus, the rear suspension tends to compress or squat. Driving forces of the vehicle such as a chain or belt may be utilized to counteract the tendency to squat during acceleration. This is known in vehicle dynamics as anti-squat.

(76) Anti-squat is typically described as a percentage value. 100% anti-squat is when the anti-squat force is equal and opposite to the load transfer force due to acceleration. As a result, the system is in equilibrium and no suspension squat occurs. Over 100% anti-squat is when the anti-squat force is both opposite and greater than the load transfer force and therefore the suspension extends during acceleration. Anti-squat of 0% is when there is zero anti-squat force to counteract the load transfer and therefore suspension squats during acceleration. Anti-squat between 0-100% is when the anti-squat force is both opposite and less than the load transfer force and therefore the suspension squats during acceleration but to a lesser degree than with 0% anti-squat. A negative anti-squat percentage is when the anti-squat force acts in the same direction on the rear wheel as the load transfer force and therefore the squat due to load transfer is magnified. Anti-squat is directly related to the DIVC[C] migration of the suspension linkage. Anti-squat around or slightly above 100% is ideal where pedaling occurs typically around the first half of travel to improve pedaling efficiency. After this point, an anti-squat below 100% is preferred so that the driving force is reduced, and the suspension can be utilized later in the travel where pedaling typically does not occur. This may also reduce feedback from the driving force to the rider. Too high of an anti-squat is less than ideal because it results in high feedback from the driving force to the rider and is detrimental to pedaling efficiency because the load transfer and anti-squat force are far from equilibrium.

(77) When the vehicle decelerates, the force distribution changes and the front wheel load is increased while the rear wheel load is decreased. As a result, the rear suspension tends to extend or rise. This is known in vehicle dynamics as anti-rise. The magnitude of anti-rise is directly related to the DIVC[E] migration.

(78) 100% anti-rise is when the anti-rise force is equal and opposite to the load transfer force due to deceleration. As a result, the system is in equilibrium and no suspension rise occurs. Over 100% anti-rise is when the anti-rise force is both opposite and greater than the load transfer force and therefore the suspension squats during deceleration. Anti-rise of 0% is when there is zero anti-rise

force to counteract the load transfer and therefore suspension rises during deceleration. Anti-rise between 0-100% is when the anti-rise force is both opposite and less than the load transfer force and therefore the suspension rises during deceleration but to a lesser degree than with 0% anti-rise. A negative anti-rise percentage is when the anti-rise force acts in the same direction on the rear wheel as the load transfer force and therefore the rise due to load transfer is magnified. Anti-rise less than 100% may help improve traction while anti-rise greater than 0% may help stabilize geometry during deceleration. Therefore, an anti-rise in the 50-100% can be a suitable range for an improved ride.

(79) Another suspension characteristic is the leverage rate (LR), which is the ratio of the change in vertical wheel travel to the change in shock stroke. A plot can be generated to represent the instantaneous leverage rate from the fully extended to the fully compressed state. The motion ratio (MR) is the inverse of the LR. Generally, the higher the leverage rate the greater the mechanical advantage on the shock/damper and the lower the force that compresses the shock. Generally, the lower the leverage rate the lesser the mechanical advantage on the shock/damper and the higher the force that compresses the shock.

(80) The direction of the DWAP is also a suspension performance metric. The ability of the rear suspension to absorb bumps when the vehicle is moving forward increases the more rearward the path from the extended state to the compressed state. The rearward path allows the rear wheel to move with the impact improving performance.

(81) Anti-squat, rate of change of the power transmitting length (dPTL), described below, anti-rise, leverage rate and DWAP direction are four performance metrics or suspension characteristics pertinent to suspension designs. Anti-squat and DWAP are directly related to the driving cog axis location. As the suspension linkage arrangement is altered so that the DWAP direction is more rearward, DIVC[A] typically is raised the anti-squat percentage will increase. This is discussed in further detail below during the discussion of the chain/belt and frame components of anti-squat. This increase may result in poor suspension performance due to excessive extension of the wheel carrier body under acceleration, stiffening the suspension and reducing traction and comfort.

(82) As the DIVC[D] is moved higher above the ground, the DWAP direction will tend to move rearward. As a result, the anti-rise percentage will increase. This increase may result in poor suspension performance due to excessive compression of the wheel carrier body under deceleration, stiffening the suspension and reducing traction and comfort.

(83) In order to manipulate the magnitudes of the anti-squat and/or anti-rise percentages and simultaneously increase the rearward direction of the DWAP, a drive train vector adjustment mechanism may be utilized with a 6-bar linkage. In one example, the drive train vector adjustment mechanism may be a drive train idler positioned between the driving portion of the drive train and the driven portion of the drive train. In a more particular example, the idler may be an idler cog that is located between the driving cog and the driven cog. This idler cog may change the chain force vector angle. Changing this chain force vector angle can manipulate the anti-squat percentage. For example, incorporating the idler cog may reduce the anti-squat percentage as compared to a system that does not include an idler cog. Additionally, or alternatively, this 6-bar layout also allows the DIVC[D] to be located further forward than conventional designs allowing for lower anti-rise percentages. These concepts and further derivations are discussed below. The various embodiments discussed below are discussed using an idler cog. It is however understood that other vector adjustment mechanisms can also be used in place of the idler cog for the respective systems. For example, simple pulleys may be used for belt driven systems or cable driven systems. In some examples, a pin or other vector adjustment device may be used to offset the drive mechanism, so the force vector of the drive mechanism is moved away from the driving member (e.g. the driven gear, cog, etc.)

(84) Based upon the number of bodies present in the structure, the total number of instantaneous velocity centers (IVCs) can be determined. The following equation can be used:

(85) $\text{NumberofInstanceCenters} = N = \frac{n(n-1)}{2}n = \text{numberofbodiesmovingrelativetooneanother}$
 $N = \text{totalnumberofinstantaneousvelocitycentersofthelinkage}$

(86) As an example, this equation may be applied to a 4-bar linkage. In this example, $n=4$. The following solves the equation for a 4-bar linkage:

(87) $N_4 = \frac{4(4-1)}{2} = \frac{12}{2} = 6$

(88) This example shows that there are 6 total instantaneous velocity centers for a 4-bar linkage.

(89) As another example, this equation may be applied to a 6-bar linkage. In this example, $n=6$. The following solves the equation for a 6-bar linkage:

(90) $N_6 = \frac{6(6-1)}{2} = \frac{30}{2} = 15$

(91) This example shows that there are 15 total instantaneous velocity centers for a 6-bar linkage.

(92) In accordance with various embodiments, the suspension system can include a suspension setup having more than four links. It may be noted that while some of the concepts discussed herein might be accomplished with four links, in some of the embodiments discussed herein, as shown herein by example, six links are used. As is suitable, more or few links can also be used to accomplish the various concepts as discussed herein.

(93) As noted above, there are 15 IVCs in a 6-bar linkage.

(94) $N_6 = \frac{6(6-1)}{2} = \frac{30}{2} = 15$

(95) A selection of the 15 total IVCs can be determined visually without being derived using other IVCs as further described below. As used herein, these IVCs are known as physical IVCs, or PIVCs. PIVCs are defined at the pivotal axes or virtual pivotal axes of jointed linkage body members. There are four PIVCs in a 4-bar linkage while there are seven PIVCs in a 6-bar linkage.

(96) FIG. 1A shows a bicycle frame having an example of a 6-bar suspension. This drive train layout has a direct drive from the drive cog to the driven cog without an idler in-between. Here, crank arm 54 and driving cog 55 rotate about driving cog axis 19. The suspension includes a suspended body 1, a link body 2, a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled with one another. Link body 2 is the wheel carrier and the brake carrier body in this embodiment. In other embodiments, the wheel carrier and the brake carrier body may differ. The link body 3 includes jointed connections with the suspended body 1 defining PIVC[1][3] 7, the link body 4 defining PIVC[3][4] 9, and the link body 6 defining PIVC[3][6] 13. The link body 4 includes an additional jointed connection with the link body 2 defining PIVC[2][4] 10. The link body 5 includes additional jointed connections with suspended body 1 defining PIVC[1][5] 8, link body 2 defining PIVC[2][5] 11, and the link body 6 defining PIVC[5][6] 12. The suspension includes a damper unit 16 pivotally connected to link body 5 at joint 17 and suspended body 1 at joint 18. The damper unit 16 is configured to resist movement between the two bodies. FIG. 1A also illustrates the brake caliper mounts 30 and rear wheel axis positioned in the extended state 14-E. Both the brake caliper mounts 30 and the rear wheel axis are also shown as being located on link body 2. Link body 2 may be referred to the seatstay in this embodiment, however the seatstay may be a different link body in other embodiments. Link body 4 may be referred to the chainstay in this embodiment, however the chainstay may be a different link body in other embodiments. Also shown is offset 600, which is the distance between the driving cog axis and PIVC[1][3] 7. This offset is pertinent to and manipulative of the magnitude of rearward direction of DWAP 100.

(97) FIG. 1B shows a portion of the embodiment of FIG. 1A with the location of cross section C-C denoted. Here the following are shown: Driving cog axis 19; driving cog 55; chain/belt 57; ground 61; front driving cog plane 1200 which is parallel to driving cog axis 19, perpendicular to the ground, and tangent to the diameter of driving cog 55; rear driving cog plane 1201 which is parallel to driving cog axis 19, perpendicular to the ground, and tangent to the diameter of driving cog 55; driving cog region width 1202 which is equivalent to the driving cog diameter; driving cog radius 1600 which is equivalent to half of the driving cog diameter; and cross section line C. Cross section line C is located within the region defined by the ground 61, front driving cog plane 1200, and rear

driving cog plane **1201**.

(98) FIG. 1C shows cross-section C-C of the embodiment of FIG. 1B. Here the following are shown: Driving cog axis **19**; driving cog **55**; driving cog center plane **1400**; chain/belt **57**; crank arm assembly **54**; suspended body-1; link body-3; link body-6; link body-4; point **1500** located on link body-4; link body-4 reference plane **1501** that is normal to driving cog axis **19** and coincident to point **1500**; frame center plane **1300**; chain/belt-line **1401** which is the distance from frame center plane **1300** to driving cog center plane **1400**; and link body-4 reference distance **1502** which is the distance from frame center plane **1300** to link body-4 reference plane **1501**. Here it is clear that chain/belt-line distance **1401** is greater than link body-4 reference distance **1502**. Therefore, no portion of link body-4 is located outside (“outside” being the direction, relative to the plane, away from the vehicle frame/suspension) of driving cog center plane **1400**.

(99) In accordance with various embodiments, the suspension system includes a 6-bar linkage and an idler cog. FIG. 2A shows a bicycle frame with a 6-bar suspension having an idler cog **56**. Note, an example of the idler is not shown in FIG. 2A for clarity. The idler cog **56** is shown in FIG. 2B. As discussed above the idler is shown as an idler cog as used with a chain drive. As illustrated, the crank arm **54** and driving cog **55** rotate about driving cog axis **19**. The suspension includes a suspended body **1**, a link body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled with one another. In this embodiment, the link body **3** includes jointed connections with the suspended body **1** defining PIVC[1][3] **7**, the link body **4** defining PIVC[3][4] **9**, and the link body **6** defining PIVC[3][6] **13**. The link body **4** includes an additional jointed connection with the link body **2** defining PIVC[2][4] **10**. The link body **5** includes additional jointed connections with suspended body **1** defining PIVC[1][5] **8**, link body **2** defining PIVC[2][5] **11**, and the link body **6** defining PIVC[5][6] **12**. The suspension includes a damper unit **16** pivotally connected to link body **5** at **17** and suspended body **1** at **18** configured to resist movement between the two bodies. Brake caliper mounts **30** and rear wheel axis at the extended state **14-E** are located on link body **2** which is also the wheel carrier and brake carrier body in this embodiment.

(100) FIG. 2A also illustrates the PIVC[1][3] **7** and offset **601**. The offset **601** is the distance between the driving cog axis and PIVC[1][3] **7**. This distance is pertinent and manipulative of the magnitude of rearward direction of DWAP **15**. Note that offset **601** (of FIG. 2A) is greater than offset **600** (FIG. 1A), and as a result, DWAP **15** is more rearward compared to DWAP **100**.

(101) FIG. 2B shows the bicycle frame from FIG. 2A with additional components for clarity. As illustrated, the rear wheel **50** can be pivotally connected to link body **2** and rear wheel axis **14-E**. A rear cassette **52** is a group of various driven cog sizes that chain/belt **57** can be shifted across mechanically via rear derailleur mechanism **53**. Also shown are front suspension fork **58** wherein front wheel **60** is pivotally connected at front wheel axis **59**. Rear wheel **50** and front wheel **60** contact the surface **61**. Note that the surface **61** (e.g., the ground) is depicted as a line, but represents a plane/surface in 3d-space.

(102) As shown in this particular embodiment, idler cog **56** includes an axis **75** that is coincident to IVC[1][3] **7**. In other embodiments, these axes may be different (see embodiments, below).

Chain/belt **57** is routed from driving cog **55**, which is rigidly connected to crank arm assembly **54**, to idler cog **56** to a driven cog of cassette **52** through the cogs of rear derailleur **53** and back to the driving cog **55** in a continuous loop. When the crank arm assembly **54** is rotated, power is transmitted from driving cog **55** to a driven cog of cassette **52** via the idler cog **56**. As a result, the chain force vector is along the top part of the chain/belt from the tangent connection point of the driven cog of cassette **52** to the tangent connection point of idler cog **56**. Note that the cassette **52** with multiple driven cogs and derailleur mechanism **53** may be absent in other embodiments. In other embodiments, a single driven cog may be used.

(103) FIG. 2C shows the bicycle frame from FIG. 2A in the compressed state. Here, rear wheel axis **14-C** is shown. FIG. 2D shows the bicycle frame from FIG. 2B in the compressed state. FIG. 2E shows the 6-bar suspension linkage system of FIG. 2A in the extended state with cross-section line

A-A passing through the idler cog **56**.

(104) FIG. 2F illustrates a cross sectional view taken at cross-section A-A of FIG. 2E showing details of idler cog **56** assembly. For example, link body **3** can be pivotally mounted to is suspended body **1** about PIVC[1][3] axis **7**. In various embodiments, the link body **3** includes a pivotable joint mechanism. For example, a set of bearings **200** can be pressed into link body **3** with bearing sleeve **203** in-between, and inner race extensions **201** to the exterior. While a bearing mechanism is described herein as an example, other joints can also be utilized (e.g. torsion joints). Axle **202** pins link body **3** to suspended body **1** and threads into a nut housed in suspended body **1**. Axle **202** has an expanding collet head feature so that when wedge **205** is compressed axially by bolt **206**, the head of axle **202** expands against suspended body-**1** preventing axle **202** from loosening. Bearing **300** is pressed into idler cog **56** and bolt **301** secures the idler cog and bearing to axle **202** so that both PIVC[1][3] **7** and idler cog axis **75** are coincident.

(105) When idler cog axis **75** is coincident to PIVC[1][3] **7**, the assembly can be optimized for weight since the idler assembly can be integrated into the pivot assembly of link body **3**. This also provides fewer machined interfaces simplifying manufacturing and potentially lower costs. The chain force about idler cog **56** due to the driving cog **55** can be of high magnitude and the axle **202** provides adequate strength.

(106) In accordance with various embodiments, the transverse envelope of the link body **4** is not limited by the chain ring **55**. For example, some or all of the link body **4** is positioned above the chain ring **55**. In the case of FIG. 1A, link body **4** is lower in the assembly and is positioned behind driving cog **55** in this view. As a result, the link body **4** structure must be sufficiently narrow when viewed from the top view (normal to the ground) in order to limit or prevent interference with the chainring **55** and rear wheel/tire **50**. With embodiment 1, link body **4** is located higher on the frame and is above driving cog **55**. As a result, link body **4** can be much wider from the back view which is illustrated further in FIGS. 1B-1C and 2G-2H. This allows for a stronger structure with the ability to provide additional tire clearance which allows for a greater range of tire size compatibility and a greater ability to shed mud, dirt and other trail debris.

(107) FIG. 2G shows a section of the embodiment of FIG. 2B with cross section D-D. Here the following are shown: Driving cog axis **19**; driving cog **55**; chain/belt **57**; ground **61**; front driving cog plane **1200** which is parallel to driving cog axis **19**, perpendicular to the ground, and tangent to the diameter of driving cog **55**; rear driving cog plane **1201** which is parallel to driving cog axis **19**, perpendicular to the ground, and tangent to the diameter of driving cog **55**; driving cog region width **1202** which is equivalent to the driving cog diameter; driving cog radius **1600** which is equivalent to half of the driving cog diameter; and cross section line D-D. Cross section line D-D is located within the region defined by the ground **61**, front driving cog plane **1200**, and rear driving cog plane **1201**.

(108) FIG. 2H shows cross-section D-D of the embodiment of FIG. 2G. Here the following are shown: Driving cog axis **19**; driving cog **55**; driving cog center plane **1400**; chain/belt **57**; crank arm assembly **54**; suspended body **1**; link body **3**; link body **6**; link body-**4**; point **1510** located on link body **4**; link body **4** reference plane **1511** that is normal to driving cog axis **19** and coincident to point **1510**; frame center plane **1300**; chain/belt-line **1411** which is the distance from frame center plane **1300** to driving cog center plane **1400**; and link body **4** reference distance **1512** which is the distance from frame center plane **1300** to link body **4** reference plane **1511**. Here it is clear that chain/belt-line distance **1411** is less than link body **4** reference distance **1512**. Therefore, a portion of link body **4** is located outside ("outside" being the direction, relative to the plane, away from the vehicle frame/suspension) of driving cog center plane **1400**. The ability to extend link body **4** outward of driving cog center plane **1400** increases the strength of link body **4** by allowing for a larger cross section while maintaining clearance to driving cog body **55**. It also allows for greater clearance between link body **4** and rear wheel/tire **50** which allows for more clearance for trail debris such as mud and dirt.

(109) In accordance with various embodiments, the link body **6** length is minimized with the addition of idler cog **56**. As link body **4**, and link body **3** are moved upwards and dimension of **601** is greater than that of **600**, the size of link body **6** decreases. This decreases the weight of link body **6** aiding in the overall performance of the machine.

(110) In accordance with various embodiments, the anti-squat percentage can be adjusted by increasing or decreasing the diameter of idler cog **56**. However, the efficiency of the chain/belt force transfer from idler cog **56** may decrease with decreased cog diameter. In accordance with various embodiments, alternative tuning can be performed by changing the relationship between idler cog axis **75** and PIVC[1][3] **7**. For example, idler cog axis **75** can be moved so that it is not coincident to PIVC[1][3] **7**. This alternative structure is discussed in more detail in the disclosure of FIGS. **3A-3F**.

(111) With regards to FIGS. **3A-3F** and in accordance with various embodiments, the suspension system includes a 6 bar linkage. FIG. **3A** shows a bicycle frame with a 6 bar suspension with idler cog **56** with axis **75**. Here, crank arm **54** and driving cog **55** rotate about driving cog axis **19**. The suspension includes a suspended body **1**, a link body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled with one another. In accordance with various embodiments, the link body **2** is the wheel carrier and the brake carrier body. In other embodiments, the wheel carrier and the brake carrier body may differ. The link body **3** includes jointed connections with the suspended body **1** defining PIVC[1][3] **7**, the link body **4** defining PIVC[3][4] **9**, and the link body **6** defining PIVC[3][6] **13**. The link body **4** includes an additional jointed connection with the link body **2** defining PIVC[2][4] **10**. The link body **5** includes additional jointed connections with suspended body **1** defining PIVC[1][5] **8**, link body **2** defining PIVC[2][5] **11**, and the link body **6** defining PIVC[5][6] **12**. The suspension includes a damper unit **16** pivotally connected to link body-5 at **17** and suspended body **1** at **18** configured to resist movement between the two bodies. Brake caliper mounts **30** and rear wheel axis at the extended state **14-E** are located on link body **2** which is also the wheel carrier and brake carrier body in this embodiment. In accordance with various embodiments, the link body **2** is the seatstay. Alternatively, the seatstay may be a different link body in other embodiments. In accordance with various embodiments, the link body **4** may be referred to the chainstay. Alternatively, the chainstay may be a different link body in other embodiments. Note that idler cog axis **75** is offset from (i.e. not coincident with) PIVC[1][3] **7**. Also shown is distance **601**, which is the distance between the driving cog axis and PIVC[1][3] **7**. This distance is pertinent to the magnitude of rearward direction of DWAP **15**. Note that offset **601** is greater than offset **600**, and as a result, DWAP **15** is more rearward compared to DWAP **100**.

(112) FIG. **3B** shows the bicycle frame from FIG. **3A** with additional components for clarity. Here rear wheel **50** is pivotally connected to link body **2** and rear wheel axis **14-E**. A rear cassette **52** is a group of various driven cog sizes. The chain/belt **57** can be shifted into mechanically via rear derailleur mechanism **53** moving between the various driven cogs. Idler cog **56** includes axis **75**. Axis **75** is offset from IVC[1][3] **7**. Chain/belt **57** is routed from driving cog **55**, which is rigidly connected to crank arm assembly **54**, to idler cog **56** to a driven cog of cassette **52** through the cogs of rear derailleur **53** and back to the driving cog **55** in a continuous loop. When the crank arm assembly **54** is rotated, power is transmitted from driving cog **55** to a driven cog of cassette **52** via the idler cog **56**. As a result, the chain force vector is along the top part of the chain/belt from the tangent connection point of the driven cog of cassette **52** to the tangent connection point of idler cog **56**. Note that the cassette **52** with multiple driven cogs and the derailleur mechanism **53** may be absent in other embodiments. In other embodiments, a single driven cog may be used. Also shown is a front suspension fork **58** wherein front wheel **60** is pivotally connected at front wheel axis **59**. Rear wheel **50** and front wheel **60** contact the ground **61**.

(113) FIG. **3C** shows the bicycle frame from FIG. **3A** in the compressed state. Here, rear wheel axis **14-C** is shown. FIG. **3D** shows the bicycle frame from FIG. **3B** in the compressed state. FIG. **3E** shows the 6-bar suspension linkage system of FIG. **3A** in the extended state with cross-section line

B-B through idler cog **56**.

(114) FIG. 3F shows the cross-section taken along line B-B of FIG. 3E showing details of the idler cog **56** assembly. Here bearing **300** is pressed into idler cog **56** and bolt **301** secures the idler cog and bearing to threaded insert **210** housed within suspended body-1.

(115) When idler cog axis **75** is offset from PIVC[1][3] **7**, a greater range of anti-squat percentages are possible. The idler cog diameter can remain the same, while a large change in anti-squat is possible depending upon the location of idler cog axis **75**. The anti-squat percentage can be fine-tuned by increase or decreasing the diameter of idler cog **56**.

(116) Tony Foale (Foale, Tony. *Motorcycle Handling and Chassis Design the Art and Science. Second Edition*. Spain: Tony Foale Designs by Tony Foale, 2002. PDF accessed 2011.)

incorporated herein by reference in its entirety, details a simple graphical method to determine anti-squat and anti-rise percentages by using a side view of a belt or chain-driven two-wheel vehicle.

(117) The method described in Tony Foale is used in the analysis shown in FIG. 4. Shown in FIG. 4 are the following: Driven wheel **50**; front wheel **60**; Front wheel axis **59**; Driven wheel axis at the extended state **14-E**; Driven wheel axis at an intermediate state **14-X**; Driven wheel axis at the compressed state **14-C**; Driven wheel axis path (DWAP) **15** and DWAP **100** for comparison.

Ground line **61** tangent to driven wheel at extended state and perpendicular to gravity; Tangent point **508** of front wheel **60** to ground line **61**; Ground line at an intermediate state **509** is parallel to **61**; Driven wheel tire to ground tangent point at an intermediate state **510**; Total driven wheel suspension travel distance perpendicular to the ground line known as the total vertical wheel travel **511**; Intermediate driven wheel suspension travel distance perpendicular to the ground line known as the intermediate vertical wheel travel **512**; Driving cog **55** and driving cog axis **19**; Driven cog **514**; Chain force vector **515** that is tangent to the tops of idler cog **56** and the driven cog **514**; Idler cog **56** rotates about idler cog axis **75**; DIVC[AD] at the extended state **516**; DIVC[AD] at an intermediate state **517**; DIVC[AD] at the compressed state **518**; DIVC[AD] migration path **519**; Driving force vector **520** drawn through the driven wheel axis at an intermediate state **14-X** and the DIVC[AD] at an intermediate state **517**; Instantaneous Force Center (IFC) **521** located at the intersection of chain force vector **515** and driving force vector **520**; Anti-Squat force vector **522** drawn through the driven wheel tire to ground tangent point at an intermediate state **510** and the Instantaneous Force Center (IFC) **521**; Squat layout line **523** which is perpendicular to the ground and passes through front wheel axis **59**; Anti-Squat definition point **524** where Anti-Squat force vector **522** intersects with Squat layout line **523**; Anti-Squat measured distance **525** is the perpendicular distance from the ground line **61** to the Anti-Squat definition point **524**; Anti-Rise force vector **526** is drawn through driven wheel tire to ground tangent point at an intermediate state **510** and DIVC[AD] at an intermediate state **517**; Anti-Rise definition point **527** where Anti-Rise force vector **526** intersects the Squat layout line **523**; Anti-Rise measured distance **528** is the perpendicular distance from the ground line **61** to the Anti-Rise definition point **527**; Center of Gravity (COG) **529** is the mass of the suspended body of the vehicle including the rider, passengers and any cargo; COG horizontal **530** is a line drawn parallel to the ground through COG **529**; COG definition point **531** is the point in which the COG horizontal **530** intersects the Squat layout line **523**; COG measured distance **532** is the perpendicular distance from the ground line **61** to the COG horizontal **530**.

(118) Anti-Squat may be defined as:

$$(119) \text{ Anti - Squat} = \left(\frac{\text{Anti - Squatmeasureddistance}}{\text{COGmeasureddistance}} \right) 100\%$$

Anti-Squat in this example is then equal to:

$$(120) \text{ Anti - Squat} = \left(\frac{\text{Anti - Squatmeasureddistance}(525)}{\text{COGmeasureddistance}(532)} \right) 100\%$$

Anti-Rise may be defined as:

$$(121) \text{ Anti - Rise} = \left(\frac{\text{Anti - Risemeasureddistance}}{\text{COGmeasureddistance}} \right) 100\%$$

Anti-Rise in this example is then equal to:

(122) Anti - Rise = $\left(\frac{\text{Anti - Rise measured distance}(528)}{\text{COG measured distance}(532)}\right)100\%$

(123) Anti-squat and anti-rise may be calculated at all points from the extended state to the compressed state to generate anti-squat and anti-rise curves. These curves are typically plotted as a function of “vertical wheel travel” which is equivalent to the total driven wheel suspension travel distance **511** perpendicular to the ground line **61** in FIG. **4**. The anti-squat curve will change depending upon the sizes of idler cog **56** and driven cog **514** since this will change the location of the Instantaneous Force Center (IFC) **521**. Note that in this example the DIVC[AD] is considered. As a result, both the anti-squat and anti-rise may be calculated using the DIVC[AD] migration. If the suspension linkage was arranged so that the DIVC[A] was separate from the DIVC[D], the DIVC[A] migration would be used to calculate the anti-squat, while the DIVC[D] migration would be used to calculate the anti-rise using the same methodology.

(124) When the suspended body is loaded with a rider, passenger or cargo the suspension will compress or sag to a desired vertical wheel travel at sag point between the extended and compressed state. The preferred sag point varies depending upon desired ride characteristics but typically ranges between 15-45%. The suspension will be positioned near this sag point as the vehicle accelerates from a static position.

(125) The sag percentage is defined as the following:

(126) Sag = $\left(\frac{\text{Vertical wheel travel value at sag point}}{\text{Total vertical wheel travel value}}\right)100\%$

(127) If the suspension linkage geometry/layout is altered, (number of linkage bodies, link body lengths and PIVC locations), the anti-squat, anti-rise, and leverage rate curves may vary.

Suspension performance is therefore directly related to the suspension linkage layout. For example, FIG. **5** shows two anti-squat curves as a comparison. Anti-squat curve **900** is an example from the embodiment of FIG. **2A** where an idler cog is implemented. In this example, the dimension of **601** is about 136 mm. **906** is the anti-squat percentage of the embodiment of FIG. **2A** at the extended state, while 903 is the anti-squat percentage of the embodiment of FIG. **2A** at the compressed state. Anti-squat curve **800** is that of linkage layout of the embodiment of FIG. **1A** where no idler cog is used. In this example, dimension **600** is ~70 mm. In this case the chain force vector is tangent to the tops of driving cog **55** and the driven cog **514**. **806** is the anti-squat percentage of at the extended state, while 803 is the anti-squat percentage at the compressed state in this case. The addition of the idler cog of the embodiment of FIG. **2A** greatly reduces the anti-squat percentage, but the general curve form may remain. The same may be true with the embodiment of FIG. **3B**. Anti-squat curves **600** and **800** have a generally stable and higher anti squat values initially creating an efficient pedaling platform that prevents bobbing when accelerating. This results in efficient power transfer during acceleration since energy is not being wasted to compress the shock/damper. There is then a non-linear drop-off towards the compressed state. This is beneficial because continuing a similar high anti-squat percentage in this portion of the travel since it would inhibit suspension compression from absorbing impacts.

(128) FIG. **19** depicts expected/exemplary anti-squat curves of a 6-bar suspension linkage system of the present disclosure with relatively smaller and larger driven mechanisms (e.g., differently sized cogs of a rear cassette). For example, as the chain/belt **57** is moved between differently sized driven mechanisms (e.g., between differently sized cogs **514** of a rear cassette **52**), the anti-squat behavior of the suspension linkage may change. For example, when the belt/chain **57** drives a relatively larger cog **514** (i.e., the bike is in a relatively lower gear), the anti-squat may initially increase as the vertical wheel travel increases (e.g., between about 0-mm and about 35-mm). As the vertical wheel travel increases further, the anti-squat may fall. The relationship between anti-squat and vertical wheel travel may have a maximum point between the minimum and maximum vertical wheel travel. When the belt/chain **57** drives a relatively smaller cog **514** (i.e., the bike is in a relatively higher gear), the anti-squat may fall between the minimum and maximum vertical wheel travel. The anti-squat may have a maximum value at a minimum value of vertical wheel travel and

a minimum value at a maximum value of vertical wheel travel. Other driven mechanisms may exhibit other suitable anti-squat behavior.

(129) FIG. 20 is a graph depicting an expected/exemplary migration path 533 of an IFC 521 of any suspension linkage of the present disclosure. The migration path is shown relative to the driving cog axis 19 and the ground line 61, for reference. As the suspension linkage moves between an extended and compressed state, the IFC 521 may move rearward to an intermediate position 536 and then forward to compressed position 534. For example, when a suspension linkage is in an extended state, the IFC 521 may be in an extended position 538. As the suspension linkage is compressed, the IFC may move rearward along the migration path 533 to an intermediate position 536. As the suspension is further compressed, the IFC 521 may move forward along the migration path 533 to a compressed position 534. The compressed position 534 may be forward of one or both of the extended position 538 and/or the intermediate position 536. This reversing motion of the IFC along the path 522 influences the increasing/decreasing behavior of the anti-squat in certain driven mechanism (e.g., the anti-squat of the relatively larger driven mechanism in FIG. 19).

(130) The anti-squat includes (e.g., is the sum of) two components, often referred to as “frame” anti-squat and “chain/belt” anti-squat. Chain/belt anti-squat is related to the lengthening of the Power Transmitting (PT) portion of the chain/belt (e.g., the top portion of the chain/belt 57 between to the tops of idler cog 56 and the driven cog 514), as the suspension is moved from the extended to the compressed state. Note that the lengthening of the PT portion of the chain/belt 57 may result in counter clockwise (“CCW”) rotation of the crank arm 54 when viewed as in FIG. 1A. This CCW rotation is opposite of the clockwise (“CW”) direction when the rider inputs a force on the pedals during acceleration and can be detrimental to suspension performance.

(131) The amount of lengthening of the power transmitting portion may vary depending on the diameter of driven cog 514 (e.g., as shown and described with respect to FIG. 19), which may vary depending upon which gear is chosen within the cassette 52, chain/belt anti-squat may vary as well. Thus, a “family” of anti-squat curves per driven cog diameter may be generated.

(132) To simplify the analysis of chain/belt anti-squat behavior into one approximated parameter, the Power Transmitting Length “PTL” is introduced. As shown in FIGS. 21A-B, the PTL 94 is the distance between driven wheel axis 14 and idler axis 75. The diameter of the driven or driving cogs are not considered. This PTL 94 may vary as the suspension moves from the extended state to the compressed state. FIG. 21A shows PTL 94 when the suspension in the extended state, while FIG. 21B shows the PTL 94 when the suspension in the compressed state. The PTL may be computed as a function of the vertical wheel travel “VWT”.

(133) “dPTL” is the rate of change of the PTL or the slope of the $f(VWT)=PTL$ curve. Given a set of VWT and corresponding PTL data points, dPTL may be calculated as:

$$(134) \quad dPTL = \frac{PTL}{VWT}$$

dPTL may then be plotted as a function of VWT. The magnitude of the dPTL represents the chain/belt anti-squat component that has been simplified and does not consider the driven or driven cog diameters. This is useful for understanding the overall chain/belt anti-squat behavior, and to aid in tuning of this variable with respect to sag.

(135) FIG. 22 shows an example of a possible dPTL curve capable of being generated with the disclosed suspension system plotted as a function of VWT. The dPTL curve has a non-linear behavior where the dPTL first increases as the VWT increases and then decreases as the VWT is increased further. In this example, at the extended state, VWT (0 mm), the dPTL may have a value of about 0.057. The dPTL increases to a max value of about 0.065 at VWT (45 mm), and then decreases to a value of 0.017 at the compressed state or VWT (140 mm). It also may be noted that the maximum dPTL is maximum at the sag point in this example. In addition, as DWAP 15 moves from the extended state to the compressed state, link body 3 initially rotates clockwise until it reaches inflection point position 102, it then reverses direction and rotates counter clockwise until

the compressed state.

(136) This behavior of increasing and decreasing dPTL may translate to similar behavior for an anti-squat, depending upon the driven and driving cog diameters. Maximizing dPTL at the sag point may translate to an anti-squat peak near sag depending upon the driven and driving cog diameters. FIG. 19 shows two possible anti-squat curves one with a small, and one with a large driven dog diameter given the dPTL plot in FIG. 22. As discussed above, the relatively smaller driven cog diameter curve increases and then decreases similar to the dPTL, while the large driven cog diameter curve decreases throughout the VWT.

(137) The frame anti-squat component is that which would occur should the Power PT portion of the chain/belt not lengthen as the suspension moves between extended and compressed states. This is not typical, and this component is typically used to aid in the theoretical understanding of which component, (i.e., chain/belt or frame) anti-squat contributes more greatly to the anti-squat percentage. This is useful for explaining the benefits of implementing an idler 56.

(138) In a traditional chain/belt driven two-wheeled vehicle without the use of an idler, the chain/belt anti-squat component is larger than that of the frame component. This is typically done to keep the lengthening of the PT portion of the chain/belt at low levels to reduce the amount of CCW crank arm rotation described above. This is achieved by keeping the DIVC[A] migration relatively low, and therefore resulting in a DWAP that is less rearward. As the DIVC[A] migration is raised, the frame anti-squat typically increases, the chain/belt anti-squat typically increases and the DWAP becomes more rearward. The lengthening of the PT portion of the chain/belt also increases keeping the driven and driving cogs constant.

(139) An idler 56 may be introduced to achieve the optimum anti-squat and magnitude of the lengthening of the PT portion of the chain/belt while simultaneously tuning the rearward magnitude of the DWAP. The suspension linkages of the present disclosure allow even further variable independence of anti-squat, anti-rise, dPTL, and/or leverage rate compared to existing designs. For example, the 15 IVCs that can be manipulated vs the 6 IVCs with a traditional 4-bar system. In addition, within the 6-bar layout, relationships of the IVC migrations discussed allow for fine-tuning and manipulation of these dynamic performance variables to achieve specific and improved dynamic responses.

(140) FIG. 6 shows a possible anti-rise curve 700 using this embodiment where 701 is the extended state of the suspension and 702 is the compressed state of the suspension. The anti-squat remains around 100% range which is ideal. Anti-rise less than 100% may help improve traction while anti-rise greater than 0% may help stabilize geometry during deceleration.

(141) FIG. 7 shows a possible leverage rate curve 1000 using this embodiment where 901 is the extended state of the suspension and 1002 is the compressed state of the suspension. The LR falls generally linearly from 1001 to 1002. This is preferable because the higher LR in the beginning of the travel helps improve small bump sensitivity, and the lower leverage rate at the end of the travel helps prevent harsh bottom outs. In addition, the general linear trend of the LR curve provides a supported mid-stroke and the aids in shock tuning as there are no dramatic changes in the LR.

(142) Note that anti-squat, anti-rise, leverage rate and DWAP direction are typically dependent variables in a typical 4-bar linkage or other suspension designs. As a result, the behavior of these three variables is limited with these designs. The disclosed 6-bar linkage with the addition of an idler cog 56 allows for greater separation of these variables so that each can be adjusted or optimized as discussed above to improve the ride quality.

(143) The addition of idler cog 56 to the 6-bar linkage configuration allows for both the anti-squat and anti-rise percentages to be more finely controlled resulting in improved magnitudes, while simultaneously manipulating the rearward direction of the DWAP.

(144) FIGS. 8A-8H illustrate an example of an electric bicycle, or e-bike frame including the idler 56 as described with respect to the suspension linkage of FIGS. 2A-2H. As used herein, an e-bike means any two-wheeled vehicle powered at least in part by a motive power source. For example in

the embodiment shown in FIGS. 8A-8H, the idler cog 56 axis 75 is coincident with the PIVC[1][3] 7. The suspension linkage and idler cog 56 of FIGS. 8A-8H may be substantially similar to those of FIGS. 2A-2H, applied to an e-bike frame. Further description of the suspension linkage of FIGS. 8A-8H is therefore omitted for brevity. An advantage of using an idler cog 56 with an e-bike may be that such an e-bike may resemble a pedal bike, which may have marketing appeal. Without an idler cog 56, an e-bike may look very different than a pedal bike due to packaging constraints of an e-bike. Such a different bike may be disfavored by buyers.

(145) As best shown in FIGS. 8A, 8B and 8D, the suspended body 1 of FIG. 8A includes a motive power source compartment 422 adapted to receive a motive power source 400. A motive power source 400 may be such as an electric motor powered by an on-board power source such as a battery. In other examples, a motive power source 400 may be an engine such as an internal combustion engine, or a combination of an engine and a motor, or a combination of one or more of these with a transmission. In some examples, a gearbox may be coupled to the suspended body-1 and may be used with any idler disclosed herein. The motive power source 400 may be coupled to the suspended body-1 with one or more mounts 402. An e-bike may also include a crank 54 and pedals to enable the e-bike to be powered at least in part by a rider's legs. For example, as shown in FIG. 8C, the motive power source 400 may include a crank interface 404 to allow a crank and/or driving cog 55 to couple to the motive power source 400. The motive power source 400 and/or the rider's legs, via the cranks, may power the two-wheeled vehicle. The crank interface may be any suitable interface that can transmit torque between the motive power source 400 and a crank 54. For example, the crank interface 404 may be a shaft with a spline, flat, key, threaded aperture, or other suitable structure to receive a crank 54.

(146) The suspended body-1 may include an energy storage compartment 420 adapted to receive an energy storage module 416, such as a battery. In the embodiment shown, the energy storage compartment 420 is formed in the downtube of the suspended body 1. In other embodiments, one or more energy storage compartments 420 may be formed in other portions of the suspended body 1 such as the seat tube or a top tube. The energy storage compartment 420 may be selectively closed by a panel 418, for example to enable removal of the energy storage module 416 from the energy storage compartment 420 such as for charging, replacement, and/or removal to reduce weight of the vehicle if the e-bike is to be used in a pedal-only mode, or the like. The panel 418 may be formed of a thin shell including one or more attachment mechanisms suitable to selectively couple the panel 418 to the suspended body 1 to seal the energy storage compartment 420. The panel 418 may help prevent the ingress of dirt, water, mud, dust or other contaminants into the energy storage compartment 420. The panel 418 may contain the energy storage module 416 within the energy storage compartment 420.

(147) The energy storage module 416 may be any suitable device that can store and/or convert energy into electrical energy for use by the motive power source 400. For example, the energy storage module 416 may be a primary (e.g., single use) or secondary (e.g., rechargeable) battery. Any suitable battery technology may be used. In many implementations, the energy storage module 416 is a rechargeable battery using a technology such as lithium ion, nickel metal hydride, nickel cadmium, lead acid, or other rechargeable batteries. In some implementations, the energy storage module 416 is a single use battery, such as an alkaline battery.

(148) The energy storage module 416 is in electrical communication with the motive power source 400 via an electrical conduit 414. The electrical conduit 414 may provide electrical power to the motive power source 400. In some implementations where the motive power source 400 also generates power (e.g., regenerative braking), the energy storage module 416 may receive power from the motive power source 400 via the electrical conduit 414.

(149) As shown best in FIG. 8C, the driving cog 55 axis 19 may define a driving cog axis plane 412. The driving cog axis plane 412 may extend vertically through the driving cog axis 19. An IVC plane may extend vertically through any of the IVCs disclosed herein, such as an IVC plane 406

that extends vertically through the IVC[1][5] **8**. Similarly, an IVC plane **408** may extend vertically through the IVC[1][3] **7**. The IVC plane **408** may also extend through the idler cog **56** axis **75**, such as in the embodiment shown in FIG. **8C** where the idler cog **56** axis **75** is coincident with the IVC[1][3] **7**. In many embodiments, the IVC plane **406** may be disposed forward of the IVC plane **408**. In many embodiments, the IVC plane **408** may be disposed close to the driving cog axis plane **412**. Disposing the IVC plane **408** close to the driving cog axis plane **412** may have the advantage of enabling a larger contact angle of the chain/belt **57** with the idler **56** than an arrangement where the planes **412** and **408** are relatively farther from one another, such as when the IVC plane **408** is disposed forward of the plane **412**. See, e.g., FIG. **8I** showing a contact angle **A** of the chain/belt **57** relative to the driving cog **55** and the idler **56**. The contact angle **A** is formed between a forward vertical tangent **62** of the driving cog **55** and line **64** that is tangent to both the driving cog and the idler. As the IVC plane **408** is moved closer to the driving cog axis plane **412**, the contact angle **A** increases. Typically the contact angle **A** is greater than zero degrees. Larger contact angles **A** of the chain/belt **57** with the idler **56** may increase the efficiency of the energy transmission around the idler, decrease wear, decrease the likelihood of the chain/belt **57** coming loose or slipping off the driving cog **55** and/or idler **56**, and/or decrease stress on the idler **56** and/or chain/belt **57**. In general there is a balance between weight and efficiency with the diameter or tooth count of the idler **56**. A greater diameter or tooth count can better retain the chain/belt **57** with a greater chain/tooth interface. Generally the larger the diameter of the idler (i.e., greater tooth count), the lower the anti-squat of the suspension linkage and vice versa. A greater diameter or tooth count improves power transmission efficiency due to the relative chain/belt **57** angle reduction. Generally, the larger the idler **56**, the heavier and the more difficult it is to package. Thus, disposing the IVC plane **408** forward of the driving cog axis plane **412** may improve idler cog **56** efficiency without the necessity of adding a larger, heavier idler. In some embodiments, a larger idler may be used with an e-bike than with a pedal bike since weight may be less critical in an e-bike application.

(150) Also shown in FIG. **8C**, is a longitudinal axis **410** of the link body-**6**. The longitudinal axis **410** extends between the IVC[3][6] **13** and the IVC[5][6] **12**. As shown in FIGS. **8F-8H**, the longitudinal axis **410** may move from a position behind the IVC[1][5] **8** when the suspension linkage is in an at least partially extended state (e.g., FIG. **8F**) to a position over the IVC[1][5] **8** as the suspension linkage is in an intermediate state (e.g., at an inflection point shown in FIG. **8G**), and to a position in front of the IVC[1][5] **8** as the suspension linkage is moved to an at least partially compressed state (e.g., as shown in FIG. **8H**).

(151) The suspension linkage and idler cog **56** of FIGS. **9A-9H** may be substantially similar to those of FIGS. **3A-3F**, but applied to an e-bike frame. Duplicative description of the suspension linkage of FIGS. **9A-9H** is therefore omitted for brevity. Certain aspects of the embodiment of FIGS. **9A-9G** may be similar to aspects of the embodiment of FIGS. **8A-8H**, such as the aspects related to e-bike features. Such similar features are not described further, for the sake of brevity. The frame of FIGS. **9A-9G** includes an idler cog **56** whose axis **75** is not coincident with the IVC[1][3] **7**. For example, as shown in FIGS. **9A-9G**, the idler cog **56** is disposed on the suspended body-**1** at a location below the IVC[1][3] **7**. In other embodiments, the idler cog **56** and its axis **75** may be located in other suitable positions on the suspended body-**1**. The relative position of the idler cog axis **75** to the driving cog axis **19** may influence the anti-squat behavior of the suspension linkage. An advantage of the embodiment of FIGS. **9A-9G**, where the idler cog axis is not coincident with the IVC[1][3] **7**, includes fine tuning of the anti-squat performance of the suspension linkage, as such embodiments may have more flexibility of placement of the idler cog axis **75** when the idler cog axis **75** location is not coupled to the location of the IVC[1][3] **7** (e.g., as compared to the embodiment of FIGS. **8A-8H**). For example, an idler cog axis **75** position that is further away from the driving cog axis **19** (e.g., higher or closer to the IVC[1][3] **7**) reduces anti-squat of the suspension relative to a suspension whose idler cog axis is closer to the driving cog

axis **19**. Generally, as the idler cog axis **75** position gets closer to the IVC[1][3], anti-squat is reduced. For example, the position of the idler cog axis **75** shown in FIGS. **9A-9G**, may reduce the anti-squat performance of the suspension linkage keeping it closer to 100% in magnitude at sag/inflection (e.g., in the position of the suspension linkage shown in FIG. **9F**). Such a position of the idler cog axis **75** may be beneficial for an e-bike as less anti-squat may be needed with the addition of power from the motive power source **400**.

(152) FIGS. **10-13** show an embodiment of a suspended body-**1** suitable for use with the e-bike frame of FIGS. **9A-9G**. In this embodiment, the location of the idler cog axis **75** may be determined by an aperture **434** formed in the suspended body-**1**. In some embodiments, more than one aperture **434** may be formed in different locations in the suspended body-**1** and/or the aperture **434** may be a slot, such that the idler cog **56** may be selectively located in different locations in the suspended body-**1**, such as to tune anti-squat performance of the suspension linkage. In the embodiment shown in FIGS. **10-13**, the idler cog **56** may be secured to the suspended body-**1** by a washer **428** and a fastener **424**. The fastener **424** may be a screw, a bolt, or the like. The fastener **424** may be received in an internal race **440** of a bearing assembly **426**. The outer race **442** of the bearing assembly **426** may be received in the idler cog **56**, such as to allow the idler cog **56** to rotate with respect to the suspended body-**1**. The fastener **424** may be selectively couplable to a nut **430**. For example, the fastener **424** may include threads that couple with threads in the nut **430**. The nut **430** may include a keyed portion **436**. The keyed portion **436** may be selectively receivable in a receptacle **438** formed in the suspended body-**1**. The interface of the receptacle **438** and the keyed portion **436** may prevent the nut **430** from rotating relative to the suspended body-**1**. In some examples, the aperture **434** may include threads adapted to receive threads of the fastener **424** alternately to the use of the nut **430**.

(153) FIG. **14** illustrates a selected portion of the 15 IVC migration paths of the embodiments shown herein. Note that not all IVCs or IVC migrations are shown for clarity of the figure. As shown in FIG. **14**, the various parts and IVCs of the suspension system may be located at different positions in the system depending on the state of the system. For example, the driven wheel axis may be located at different positions along the driven wheel axis migration path (DWAP) **147**, which has a path length DWAP [L]. For example, the driven wheel axis **145** may be at extended state position as shown in FIG. **14**, at compressed state position **146** as shown by the termination of DWAP **147**, or at any other position along the DWAP **147**. As another example, IVC[115][117] **133** may be located at different positions along the IVC[115][117] migration path **150**. For example, IVC[115][117] may be at extended state position **148**, at compressed state position **149**, or at any other position along the IVC[115][117] migration path **150**. DIVC[AD][113][114] may be located at different positions along the DIVC[AD][113][114] migration path **156**, which has a path length DIVC[AD][113][114][L]. For example, DIVC[AD][113][114] may be at extended state position **154**, at compressed state position **155**, or at any other position along the DIVC[AD][113][114] migration path **156**. IVC[115][118] may be located at different positions along the IVC[115][118] migration path **160**. For example, IVC[115][118] may be at extended state position **157**, at inflection point **158** of bottom link **115**, at compressed state position **159**, or at any other position along the IVC[115][118] migration path **160**. In other words, as DWAP moves from the extended state to the compressed state, link body **115** initially rotates clockwise until IVC[115][118] reaches inflection point position **158**, then reverses direction and rotates counter clockwise until IVC[115][118] reaches the compressed position **159**. IVC[113][118] may be located at different positions along the IVC[113][118] migration path **163**. For example, IVC[113][118] may be at extended state position **161**, at compressed state position **162**, or at any other position along the IVC[113][118] migration path **163**.

(154) The various migration paths may be composed of more than one migration path. For example, IVC[114][115] may be located at different positions along the IVC[114][115] migration path **153**, which has an inflection point and is therefore composed of two migration paths.

IVC[114][115] may be at extended state position **151**, at compressed state position **152**, or at any other position along the IVC[114][115] migration path **153**. FIG. **14** shows IVC[114][117] at the extended state **179** and SIVC[113][115] at the extended state **180**, but for the sake of clarity in the FIG. **14** does not show their migration paths.

(155) Several different relationships may exist between IVCs, as shown in FIG. **14**. For example, IVC[114][116] may have different positional relationships relative to various IVCs. IVC[114][116] may be located at different positions along the IVC[114][116] migration path **172**. IVC[114][116] may be at extended state position **170**, at compressed state position **171**, or at any other position along the IVC[114][116] migration path **172**. The relationship between IVC[114][116] at extended state position **170** and DIVC[AD][113][114] at extended state position **154** is represented by extended IVC line IVC[E] **173**. In various examples, IVC[114][116] extended tangent direction IVC[114][116][ET] **174** is coincident to IVC[114][116] at extended state position **170** and perpendicular to IVC[E] **173**. In various examples, the relationship between IVC[114][116] at compressed state position **171** and DIVC[AD][113][114] at compressed state position **155** is represented by compressed IVC line IVC[C] **175**. In various examples, IVC[114][116] compressed tangent direction IVC[114][116][CT] **176** is coincident to IVC[114][116] at compressed state position **171** and perpendicular to IVC[C] **175**.

(156) In various examples, the base line **164** extends parallel to the ground at the extended state. Rear line **165** extends perpendicular to a rear end of base line **164** and intersects with driven wheel axis **145** at extended state position. Front line **166** extends perpendicular to a front end of base line **164** and intersects with DIVC[AD][113][114] at extended state position **154**. Center line **167** is perpendicular to a center portion of base line **164** and intersects with DIVC[AD][113][114] at compressed state position **155**. The DIVC extended direction variable (DIVC[AD][EDV]) **168** extends between rear line **165** and front line **166**. The DIVC compressed direction variable (DIVC[AD][CDV]) **169** extends between rear line **165** and center line **167**.

(157) FIG. **15** is a partial exploded rear isometric of a portion of a suspension linkage according to the present disclosure. FIG. **16** shows a partial elevation view of a portion of the suspension linkage of the FIG. **15**. FIGS. **17A-C** show details of a key. FIG. **18** shows a partial section view of the portion of the suspension linkage of FIG. **16** taken along section line **18-18** of FIG. **16**. In FIGS. **15-18**, portions of the suspension linkage are removed, for clarity. As shown for example in FIG. **15-18**, the motive power source **400** may be received in the motive power source compartment **422** and secured to the suspended body **1** via the mounts **402**.

(158) As best shown in FIGS. **15** and **18**, the idler **56** may be coupled to the suspended body **1** and the suspension linkage by an axle assembly **80**. The link body **3** may be coupled to the suspended body **1** by the axle assembly **80**. The axle assembly **80** may include a fastener element **82**, a tapered collet **84**, one or more race extenders **76**, a retainer **78**, one or more bearings **70/72**, an axle **86**, a key **88**, an optional key fastener **66**, a spacer **90**, and an idler fastener **62**.

(159) With reference to FIGS. **17A-17C**, the key **88** may be formed of a body **91**. The body **91** may have a main portion **89**. The main portion **89** may have a locking shape, such as a lobe as shown. The main portion **89** may have other suitable shapes that resist rotation when coupled with the suspended body **1** or other portions of the axle assembly **80** (e.g., triangle, square, spline, star, other polygons or irregular shapes). The main portion **89** may have an aperture **81** formed therein or therethrough (e.g., may be a through hole or a blind hole). The aperture **81** may include threads **93** formed on an inner surface thereof. A shoulder **98** may extend proud of a face **92** of the main portion **89**. A wall **97** may extend from the shoulder **98** further away from the face **92** of the main portion **89**. The wall **97** may have an aperture **96** formed therein or therethrough (e.g., may be a through hole or a blind hole). As shown, in FIG. **17C**, the aperture **96** may have a first portion **96A** with a diameter that is smaller than a diameter of a second portion **96B**. The first portion **96A** and the second portion **96B** may be disposed end-to-end along the length of the aperture **96**. A step may be formed between the portions **96A**, **B** due to the different diameters of the respective portions.

Either or both of the portions **96A**, **B** may include helical threads **95**.

(160) The fastener element **82**, key fastener **66** and/or idler fastener **62** may be a variety of types of fastener. In some examples, the fastener **82** is a socket head cap screw. In some examples the key fastener is a counter sink screw. In some examples the idler fastener is a flange bolt. The fasteners **62**, **66**, and **82** include external helical fastener threads adapted to mate with other components of the assembly **80**. The tapered collet **84** has an aperture formed therein and adapted to receive the fastener **82**. The aperture **186** is sufficiently large to allow a clearance with the fastener **82** such that the fastener threads **184** do not engage the tapered collet **84** at the aperture **186**. The tapered collet **84** may be assembled with the fastener **82** and held to the fastener **82** with a retainer **78** such as a snap ring, spring, circlip, or the like. The retainer **78** may help withdraw the tapered collet **84** when disassembling the axle assembly **80**.

(161) The axle **86** may have an elongate body **87** with an expandable portion **79** at one end of the elongate body **87** and external axle threads **73** at an opposite end of the elongate body **87**. The axle **86** may optionally have internal axle threads **77** at the opposite end of the elongate body **87** from the expandable portion **79**. The axle **86** may include internal axle threads **99** in an end of the elongate body **87** near the expandable portion **79**. The inside diameter of the expandable portion **79** may be the same size or less than an outer diameter of the tapered collet **84**.

(162) The bearings **70** may allow for rotational motion of the link body **3** with respect to the axle assembly **80**. For example, the bearings **70** may be roller bearings, tapered bearings, bushings, or the like. The spacer **90** may be a hollow tube with a diameter larger than that of the axle **86**. The spacer **90** may be received over the axle **86**. The bearings **70** may be adapted to receive the race extenders **76**. The race extender **76** and the spacer **90** may stabilize and/or locate the bearings **70** in the axle assembly **80**. In some embodiments, the bearings **70** may be located or secured to the axle **86** by the spacer **90** and the race extenders **76**. For example, as shown in FIG. 17, the bearing **70** may be sandwiched between the respective ends of the spacer **90** and the respective race extenders **76**.

(163) As best shown in FIG. 17, when assembled, the key **88** may be received in an aperture **83** (see FIG. 15) formed in the suspended body **1**. The key **88** may be secured to the suspended body **1** such as by the pulling force of the engagement of the external axle threads **73** with the threads **93** in the aperture **81**, a press fit, or by other suitable structures. The key **88** may optionally be secured to the suspended body by the threaded engagement of the fastener **66** with the threads **95** of the first portion **96A** of the aperture **96**. The bearings **70** and respective race extenders **76**, and spacer **90**, may be assembled with the link body **3**, such as within an aperture of the link body **3** that is coincident with the IVC[1][3] **7**. Either of the bearings **70** may be floating or fixed along the axle **86**. For example, the link body **3** may include a locating feature, such as a step **85**, that receives an outer portion of the race extenders **76** that limits movement of the bearing **70** in a transverse direction along the axle **86**. The bearings **70** may be allowed to move along the axle **86**.

(164) The fastener element **82** may be threaded into internal threads **99** formed in the axle. As the fastener element **82** is drawn in to the axle **86**, the fastener element **82** may cause the collet **84** to expand the expandable portion **79** outward to engage the expandable portion **79** with the inner face of an aperture formed in the suspended body **1** (e.g., by friction), to secure the axle assembly **80** to the suspended body **1**.

(165) The bearing **72** may be assembled with the key **88**. For example, an inner race of the bearing **72** may be received on the wall **97** and located by the shoulder **98**. The outer race of the bearing **72** may be received in an aperture formed in the idler **56**. The idler **56** and bearing **72** may be secured to the axle assembly **80** by the threaded engagement of the fastener **62** with the threads **95** of the second portion **96B** of the aperture **96**. The structure shown in FIG. 17 may be suitable for an idler as shown in FIGS. 3A-3F and/or FIGS. 9A-9G. The PIVC[1][3] **7** may be spaced apart from the idler rotation axis **75** by a distance **D**. The distance **D** may be determined by a spacing between the aperture **81** and the aperture **96** of the key **88**. In many embodiments, the distance **D** may be less

than or equal to about 50-mm. For example, the distance D may be about 10-mm, 15-mm, 20-mm, 25-mm, 30-mm, 35-mm, 40-mm, 45-mm, or other suitable distances. In other embodiments, the idler **56** may be secured to the axle assembly **80** such as by the optional internal axle threads **77**. Such an arrangement may be suitable for an idler **56** such as shown in FIGS. 2A-2H and/or FIGS. 9A-9G.

(166) In numerous embodiments, a method of analyzing different behaviors of the suspension system is disclosed. For example, motions related to inflection point **158** of bottom link **115** may be analyzed as the suspension is moved from the extended to the compressed state. In one example, IVC[115][118] migration path **160** may be analyzed. IVC[115][118] migration path **160** reverses as the suspension moves from the extended state at least partially to the compressed state. In another example, this behavior may be visualized by analyzing IVC[113][118] migration path **163** in comparison to IVC[115][118] migration path **160**. When the suspension is in the extended state, IVC[113][118] at extended state position **161** is not coincident to IVC[115][118] migration path **160**. As the suspension moves towards the compressed state, IVC[115][118] moves from extended state position **157** towards inflection point **158** of bottom link **115** along IVC[115][118] migration path **160**. At a point between the driven wheel axis **145** at the extended state position and at the compressed state **146**, IVC[113][118] migration path **163** intersects IVC[115][118] migration path **160** at inflection point **158**. As the suspension continues to move towards the compressed state, IVC[115][118] moves from inflection point **158** towards compressed state position **159** along the IVC[115][118] migration path **160**.

(167) FIG. **14** also shows a relationship between the driven wheel axis **145** at the extended state and a driving cog axis **177**, represented by the driving wheel axis to driving cog axis (DWDC) line **178**. The seat-stay **114** is operatively coupled to a driven wheel **127**.

(168) Note that in all figures, “-E” denotes the extended state, “-C” denotes the compressed state, “-X” denotes a state between extended and compressed. These are only labeled at rear wheel axis **14** for clarity. Rear wheel axis **14-E** at the extended state, **14-C** at the compressed state, and **14-X** at a position between the extended state and the compressed state are shown in figures throughout.

(169) The above specification, examples and data provide a complete description of the structure and use of exemplary embodiments of the invention as defined in the claims. Although various embodiments of the claimed invention have been described above with a certain degree of particularity, or with reference to one or more individual embodiments, those skilled in the art could make numerous alterations to the disclosed embodiments without departing from the spirit or scope of the claimed invention. Other embodiments are therefore contemplated. For example, while some embodiments specify particular relationships between parts of the system, other relationships are contemplated. It is also contemplated that steps to any disclosed method may be performed in any order. It is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative only of particular embodiments and not limiting. Changes in detail or structure may be made without departing from the basic elements of the invention as defined in the following claims.

Claims

1. A two-wheel vehicle suspension linkage comprising: a suspended body **1**, a wheel carrier body **2**, a link body **3**, a link body **4**, a link body **5**, and a link body **6** operatively coupled defining a primary instantaneous velocity center (PIVC) at each joint between link bodies; a wheel operatively connected to the wheel carrier body **2**; a driven cog operatively connected to the wheel; a driving cog operatively connected to the suspended body **1**, the driving cog being disposed transversely inboard of at least a portion of at least one of the link bodies in a plane extending through the driving cog, parallel to gravity, and parallel to an axis of rotation of the driving cog; an idler operatively connected by an elongated flexible member to the wheel carrier body **2**, the link

body 3, the link body 4, the link body 5 or the link body 6, wherein the idler is rotatable about an idler rotation axis and wherein the idler is pivotally connected to the suspended body; the elongated flexible member is disposed along a route between the driving cog and the driven cog, wherein the idler is disposed in the route between the driving cog and the driven cog; and a damper unit configured to resist movement between two or more of the suspended body 1, wheel carrier body 2, link body 3, link body 4, link body 5, or link body 6.

2. The two-wheel vehicle suspension linkage of claim 1, wherein the idler is rotatably connected to suspended body 1 concentrically with a linkage pivot with the wheel carrier body 2, the link body 3, the link body 4, the link body 5 or the link body 6.

3. The two-wheel vehicle suspension linkage of claim 2, wherein: the link body 3 includes jointed connections with: the suspended body 1 defining a PIVC[1][3], the link body 4 defining a PIVC[3][4], and the link body 6 defining a PIVC[3][6]; and the link body 4 includes a jointed connection with the wheel carrier body 2 defining a PIVC[2][4]; the link body 5 includes jointed connections with: the suspended body 1 defining a PIVC[1][5], the wheel carrier body 2 defining a PIVC[2][5], and the link body 6 defining a PIVC[5][6]; wherein the idler rotation axis is coincident to the PIVC[1][3].

4. The two-wheel vehicle suspension linkage of claim 3, wherein the link body 6 defines a longitudinal axis, and the longitudinal axis moves from a position behind the PIVC[1][5] when the suspension linkage is in an at least partially extended state to a position in front of the PIVC[1][5] when the suspension linkage is in an at least partially compressed state.

5. The two-wheel vehicle suspension linkage of claim 3, wherein the link body-6 has a longitudinal axis extending between the PIVC[3][6] and the PIVC[5][6], wherein the longitudinal axis is in a first position behind the IVC[1][5] when the suspension linkage is in an at least partially extended state and in a second position in front of the IVC[1][5] when the suspension linkage is in a partially compressed state.

6. The two-wheel vehicle suspension linkage of claim 1, wherein the idler rotation axis is coincident to a PIVC defined by a combination of the suspended body 1 and, the wheel carrier body 2, the link body 3, the link body 4, the link body 5, or the link body 6.

7. The two-wheel vehicle suspension linkage of claim 6, wherein the idler is mounted to an axle, wherein the axle is pivotally connecting the wheel carrier body 2, linkage body 3, linkage body 4, linkage body 5, or linkage body 6 to suspended body-1.

8. The two-wheel vehicle suspension linkage of claim 1, wherein the link body 3 includes jointed connections with: the suspended body 1 defining a PIVC[1][3], the link body 4 defining a PIVC[3][4], and the link body 6 defining a PIVC[3][6]; the link body 4 includes a jointed connection with wheel carrier body 2 defining a PIVC[2][4]; the link body 5 includes jointed connections with: the suspended body 1 defining a PIVC[1][5], the wheel carrier body 2 defining a PIVC[2][5], and the link body 6 defining a PIVC[5][6], wherein the idler rotation axis is not coincident to the PIVC[1][3].

9. The two-wheel vehicle suspension linkage of claim 8, wherein the PIVC[1][3][7] is spaced apart from the idler rotation axis by a distance of less than or equal to 50-mm.

10. The two-wheel vehicle suspension linkage of claim 1, wherein the idler comprises an idler cog.

11. The two-wheel vehicle suspension linkage of claim 1, wherein at least one PIVC migration path reverses as the suspension linkage moves from an at least partially extended state to an at least partially compressed state.

12. The two-wheel vehicle suspension linkage of claim 1, wherein the suspended body 1 includes: a motive power source compartment, and an energy storage compartment selectively coverable by a panel.

13. The two-wheel vehicle suspension linkage of claim 12, further comprising: a motive power source received in the motive power source compartment; an energy storage module received in the energy storage compartment and contained by the panel; an electrical conduit in electrical

communication with the motive power source and the energy storage module and operative to supply electrical power from the energy storage module to the motive power source.

14. The two-wheel vehicle suspension linkage of claim 12, wherein the idler rotation axis is coincident with a PIVC.

15. The two-wheel vehicle suspension linkage of claim 12, wherein the idler rotation axis is not coincident with a PIVC.

16. The two-wheel vehicle suspension linkage of claim 1, wherein the driving cog is rotatable about a driving cog axis and the idler rotation axis is disposed above and forward of the driving cog axis.

17. The two-wheel vehicle suspension linkage of claim 1, wherein a contact angle formed between a forward vertical tangent of the driving cog and line that is tangent to both the driving cog and the idler is greater than zero degrees.

18. The two-wheel vehicle suspension linkage of claim 1, further comprising an instantaneous force center (IFC) located at an intersection of a force vector of the elongated flexible member and a driving force vector, wherein as the suspension linkage moves between an extended and compressed state, the IFC moves from an extended position rearward to an intermediate position and then forward to compressed position.

19. The two-wheel vehicle suspension linkage of claim 18, wherein the compressed position is forward of at least one of the extended position or the intermediate position.

20. The two-wheel vehicle suspension linkage of claim 1, wherein an anti-squat percentage of the suspension linkage increases as a vertical travel of the wheel increases to a first position, wherein as the vertical wheel travel increases further to a second position, the anti-squat percentage falls.

21. The two-wheel vehicle suspension linkage of claim 1, wherein a rate of change of a power transmitting length of the elongated flexible member (dPTL) increases as a vertical travel of the wheel increases to a first position, wherein as the vertical wheel travel increases further to a second position, the dPTL falls.

22. The two-wheel vehicle suspension linkage of claim 21, wherein the dPTL is maximized near a sag point of the two-wheel vehicle suspension linkage.

23. The two-wheel vehicle suspension linkage of claim 21, wherein the dPTL is maximized at a first value of the vertical travel of the wheel, and wherein the link body 3 has an inflection point at a second value of the vertical travel of the wheel greater than the first value.

24. A two-wheel vehicle suspension linkage comprising: a suspended body 1, and a plurality of link bodies including a wheel carrier body 2, a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled defining a primary instantaneous velocity center (PIVC) at each joint between the plurality of link bodies; a driving cog having a driving cog diameter and operatively connected to the suspended body 1, wherein the driving cog is rotatable in a driving cog center plane about a driving cog axis normal to the driving cog center plane, and a driving cog region is defined between: a front driving cog plane tangent to a front of the driving cog diameter, parallel to the driving cog axis, and parallel to gravity, and a rear driving cog plane tangent to a rear of the driving cog diameter, parallel to the driving cog axis, and parallel to gravity, wherein at least a portion of one or more of the plurality of link bodies is located inside the driving cog region and extends transversely outward past the driving cog center plane.

25. The two-wheel vehicle suspension linkage of claim 24, further comprising a damper unit configured to resist movement between two or more of the suspended body 1, the wheel carrier body 2, the link body 3, the link body 4, the link body 5, or the link body 6.

26. A two-wheel vehicle suspension linkage comprising: a suspended body 1, and a plurality of link bodies including a wheel carrier body 2, a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled defining a primary instantaneous velocity center (PIVC) at each joint between the plurality of link bodies; a driving cog operatively connected to the suspended body 1 and defining a driving cog axis of rotation, a driving cog radius, a driving cog center plane, and a

driving cog region, wherein; the driving cog region is defined between a front driving cog plane tangent to a front of the driving cog, parallel to the driving cog axis of rotation, and parallel to gravity and a rear driving cog plane tangent to a rear of the driving cog, parallel to the driving cog axis of rotation, and parallel to gravity, a first distance from a suspended body center plane to a vertical reference plane of one or more of the plurality of link bodies normal to the driving cog axis is greater than a second distance from the suspended body center plane to the driving cog center plane, and the first distance and the second distance are measured in the driving cog region; wherein the link body 3 includes jointed connections with: the suspended body 1 defining a PIVC[1][3], the link body 4 defining a PIVC[3][4], and the link body 6 defining a PIVC[3][6]; and the link body 4 includes a jointed connection with wheel carrier body 2 defining a PIVC[2][4]; the link body 5 includes jointed connections with: the suspended body 1 defining a PIVC[1][5], the wheel carrier body 2 defining PIVC[2][5], and the link body 6 defining an PIVC[5][6], wherein the driving cog radius is 40%-65% of a distance from the PIVC[1][3] to the driving cog axis of rotation.

27. The two-wheel vehicle suspension linkage of claim 26 further comprising a damper unit configured to resist movement between two or more of the suspended body 1, the wheel carrier body 2, the link body 3, the link body 4, the link body 5, or the link body 6.

28. A two-wheel vehicle suspension linkage comprising: a suspended body 1, and a plurality of link bodies including a wheel carrier body 2, a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled defining a primary instantaneous velocity center (PIVC) at each joint between the plurality of link bodies; a driving cog operatively connected to the suspended body 1 and defining a driving cog radius, a driving cog axis of rotation, and a driving cog center plane, wherein a portion of the link body 4 is located outside the driving cog center plane in a vertical cross-sectional view of the two-wheel vehicle suspension linkage parallel to the driving cog axis of rotation and between a front driving cog plane and a rear driving cog plane, the front driving cog plane being tangent to a front of the driving cog, parallel to the driving cog axis of rotation, and parallel to gravity, the rear driving cog plane being tangent to a rear of the driving cog, parallel to the driving cog axis of rotation, and parallel to gravity; wherein the link body 3 includes jointed connections with: the suspended body 1 defining a PIVC[1][3], the link body 4 defining a PIVC[3][4], and the link body 6 defining a PIVC[3][6]; and the link body 4 includes a jointed connection with wheel carrier body 2 defining a PIVC[2][4]; the link body 5 includes jointed connections with: the suspended body 1 defining a PIVC[1][5], the wheel carrier body 2 defining a PIVC[2][5], and the link body 6 defining a PIVC[5][6], wherein the driving cog radius is 40%-70% of a distance from the PIVC[3][6] to the PIVC[5][6].

29. The two-wheel vehicle suspension linkage of claim 28 further comprising a damper unit configured to resist movement between two or more of the suspended body 1, the wheel carrier body 2, the link body 3, the link body 4, the link body 5, or the link body 6.

30. A two-wheel vehicle suspension linkage comprising: a suspended body 1; a wheel carrier body 2; a wheel operatively connected to the wheel carrier body 2; a driven cog operatively connected to the wheel; a driving cog operatively connected to the suspended body 1; an idler operatively connected to the suspended body 1, wherein the idler is rotatable about an idler rotation axis; and an elongated flexible member disposed along a route between the driving cog and the driven cog, wherein the idler is disposed in the route between the driving cog and the driven cog, wherein an instantaneous force center (IFC) is located at an intersection of a force vector of the elongated flexible member and a driving force vector, wherein as the suspension linkage moves between an extended and compressed state, the IFC moves from an extended position rearward to an intermediate position and then forward to compressed position, wherein an axis of the wheel in the compressed state is rearward of an axis of the wheel in the extended state.

31. The two-wheel vehicle suspension linkage of claim 30 further comprising a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled with one another and the suspended

body 1 and wheel carrier body 2 and defining a primary instantaneous velocity center (PIVC) at each joint between link bodies.

32. The two-wheel vehicle suspension linkage of claim 30 further comprising a damper unit configured to resist movement between two or more of the suspended body 1, wheel carrier body 2, link body 3, link body 4, link body 5, or link body 6.

33. A two-wheel vehicle suspension linkage comprising: a suspended body 1; a wheel carrier body 2; a wheel operatively connected to the wheel carrier body 2; a driven cog operatively connected to the wheel; a driving cog operatively connected to the suspended body 1; an idler operatively connected to the suspended body 1, wherein the idler is rotatable about an idler rotation axis; and an elongated flexible member disposed along a route between the driving cog and the driven cog, wherein the idler is disposed in the route between the driving cog and the driven cog, wherein a rate of change of a power transmitting length of the elongated flexible member (dPTL) increases as a vertical travel of the wheel increases to a first position, wherein as the vertical wheel travel increases further to a second position, the dPTL falls, wherein a driven wheel axis path of the wheel is rearward from a minimum vertical wheel travel to a maximum vertical wheel travel.

34. The two-wheel vehicle suspension linkage of claim 33 further comprising a link body 3, a link body 4, a link body 5, and a link body 6 operatively coupled with one another and the suspended body 1 and wheel carrier body 2 and defining a primary instantaneous velocity center (PIVC) at each joint between link bodies.

35. The two-wheel vehicle suspension linkage of claim 33 further comprising a damper unit configured to resist movement between two or more of the suspended body 1, wheel carrier body 2, link body 3, link body 4, link body 5, or link body 6.
